

★ WEEK 13 · PART 1 of 3

Bayesian Inference: Beliefs Updated by Evidence

Part III Opens — Treat Parameters as Distributions, Not Numbers

Two views of probability · Bayes' theorem · posterior, likelihood, prior

Last week's calibration gave single numbers. Bayesian inference gives distributions — a full account of what we believe and how sure we are. It starts with one equation: Bayes' theorem.

1

PART 1 • THE CORE IDEA

Bayes' Theorem and the Update

Two ways to think about probability, Bayes' theorem as the rule for updating belief, the working form posterior proportional to likelihood times prior, and the shift to parameters as distributions.

The Minister's Posthumous Idea — Part 1

Reasoning backward, from evidence to cause

Thomas Bayes, an 18th-century Presbyterian minister, asked a backward question: given what we observe, what should we believe about the cause? His essay on this was published only after his death, in 1763.

The backward question

- Forward: given a cause, how likely is the evidence?
- Backward: given the evidence, how likely is the cause?
- Bayes gave a rule to reverse the direction.

Why it is powerful

- It matches how we actually learn from data.
- It updates a belief in light of new evidence.
- One theorem captures the whole process.

The Minister's Posthumous Idea — Part 2

A long eclipse, then a computational comeback

Laplace developed the idea further, but for much of the 20th century frequentist statistics eclipsed it. Bayesian methods returned only when cheap computing made their once-intractable calculations feasible.

The eclipse

- Frequentist methods dominated 20th-century statistics.
- The Bayesian calculations were simply too hard by hand.
- The idea waited, out of the mainstream, for decades.

The comeback

- Computers made the hard integrals tractable.
- Simulation methods (next week) unlocked real models.
- Bayes is now everywhere in modern data analysis.

From Point Estimates to Distributions

Where Week 12 leads

1

We calibrated to single numbers

Week 12 fit κ , θ , σ as point estimates. But a point hides how UNCERTAIN we are about each.

2

Bayes gives a distribution instead

Rather than one best value, we get a full distribution over the parameter — belief plus uncertainty, together.

3

One rule drives it all

Bayes' theorem updates a prior belief with data into a posterior. This week builds that machinery from the ground up.

Part 1 in one line: Bayes' theorem turns a prior belief plus data into a posterior distribution.

Two Ways to Think About Probability

The frequentist and the Bayesian

There are two coherent views of probability. The frequentist reads it as a long-run frequency; the Bayesian reads it as a degree of belief. The difference shapes what a parameter even IS.

Frequentist

- Probability is a long-run frequency of events.
- A parameter is a fixed, unknown constant.
- Randomness lives in the data, not the parameter.

Bayesian

- Probability is a degree of belief.
- A parameter is uncertain — described by a distribution.
- Belief updates as data arrives.

The Frequentist View

A fixed parameter, random data

The core idea

- The parameter has one true, fixed value.
- We estimate it from a sample of data.
- Repeat the sampling and the estimate varies.

What it gives

- Point estimates and confidence intervals.
- Statements about the PROCEDURE across repetitions.
- No probability attached to the parameter itself.

The Bayesian View

An uncertain parameter, a distribution of belief

The core idea

- The parameter is uncertain, described by a distribution.
- We start with a prior belief about it.
- Data updates that belief into a posterior.

What it gives

- A full distribution over the parameter.
- Direct probability statements about it.
- Uncertainty quantified honestly, not hidden.

Bayes' Theorem: The Update Rule

The single equation behind everything this week

$$P(\theta \mid \text{data}) = \frac{P(\text{data}|\theta) P(\theta)}{P(\text{data})}$$

The pieces

- The posterior: belief about the parameter after data.
- The likelihood: how probable the data is for each value.
- The prior: belief before seeing the data.

The denominator

- The evidence normalizes the result to a distribution.
- It is the probability of the data overall.
- It is also the computationally hard part.

The Four Ingredients

Prior, likelihood, evidence, posterior

What you bring in

- The prior: your belief before the data.
- The likelihood: the model linking parameter to data.
- The evidence: the overall probability of the data.

What comes out

- The posterior: your belief after the data.
- It blends the prior with what the data says.
- Everything Bayesian flows from these four.

Posterior is proportional to Likelihood times Prior

The form you will use constantly

$$\underbrace{P(\theta \mid \text{data})}_{\text{posterior}} \propto \underbrace{P(\text{data} \mid \theta)}_{\text{likelihood}} \times \underbrace{P(\theta)}_{\text{prior}}$$

Why drop the denominator

- The evidence doesn't depend on the parameter.
- So it is just a constant that rescales the result.
- The SHAPE comes from likelihood times prior.

How you use it

- Multiply the prior by the likelihood.
- Normalize at the end so it integrates to one.
- This is the everyday form of Bayes' theorem.

Reading the Working Form

Belief times evidence, rescaled

The multiplication

- Where the prior and likelihood agree, the posterior is high.
- Where either is near zero, the posterior is too.
- The two combine, they don't simply average.

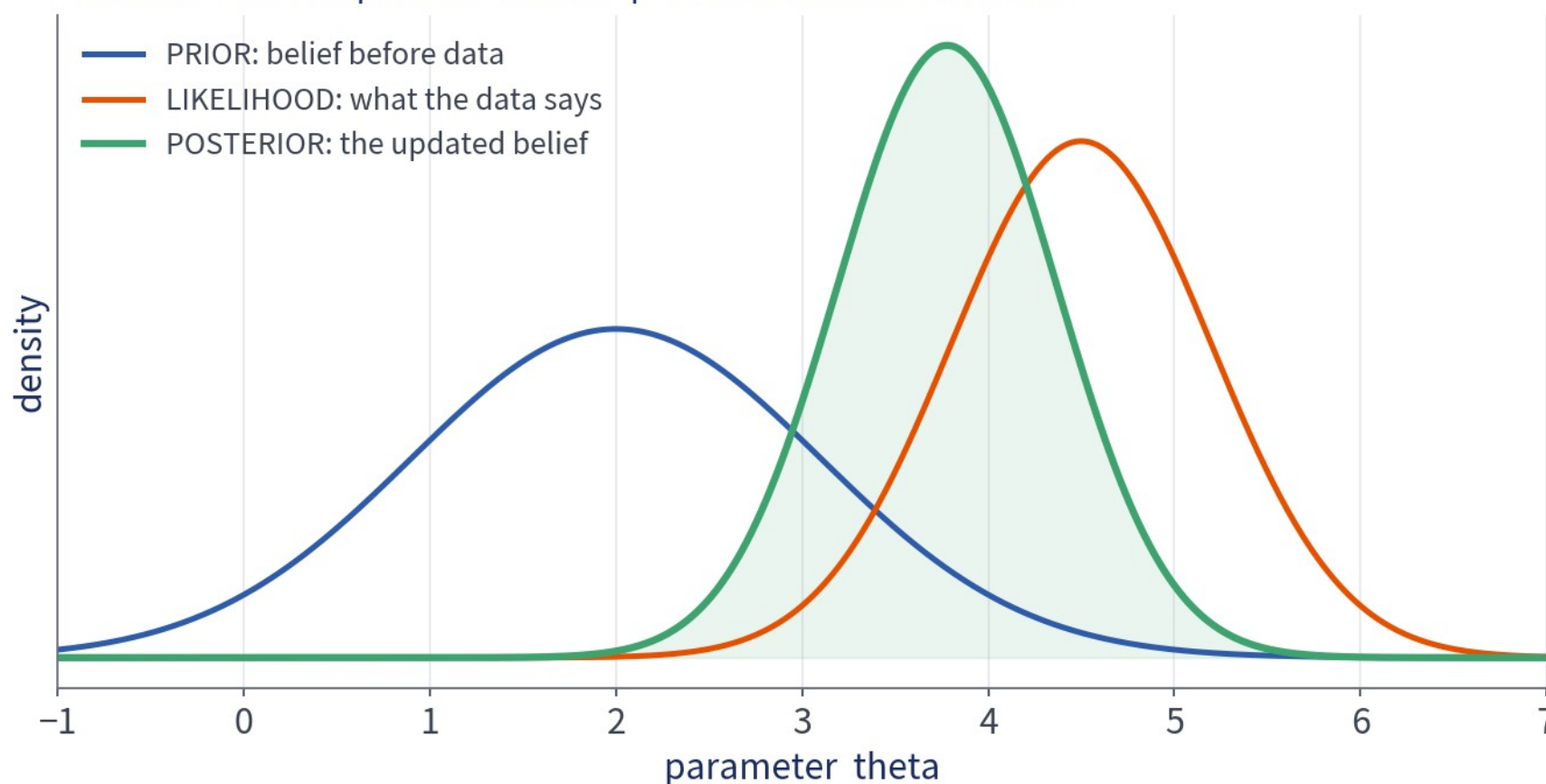
The intuition

- The prior weights values by belief.
- The likelihood weights them by fit to the data.
- The posterior keeps values that satisfy both.

Prior + Data = Posterior

The prior (blue) is belief before data; the likelihood (orange) is what the data says. The posterior (green) sits between them and is tighter than either — the compromise of belief and evidence.

Posterior = the compromise between prior belief and the evidence



The Update as a Compromise

How far the data moves your belief

What sets the balance

- A confident (narrow) prior resists the data.
- A diffuse (wide) prior lets the data dominate.
- More data pulls the posterior toward the likelihood.

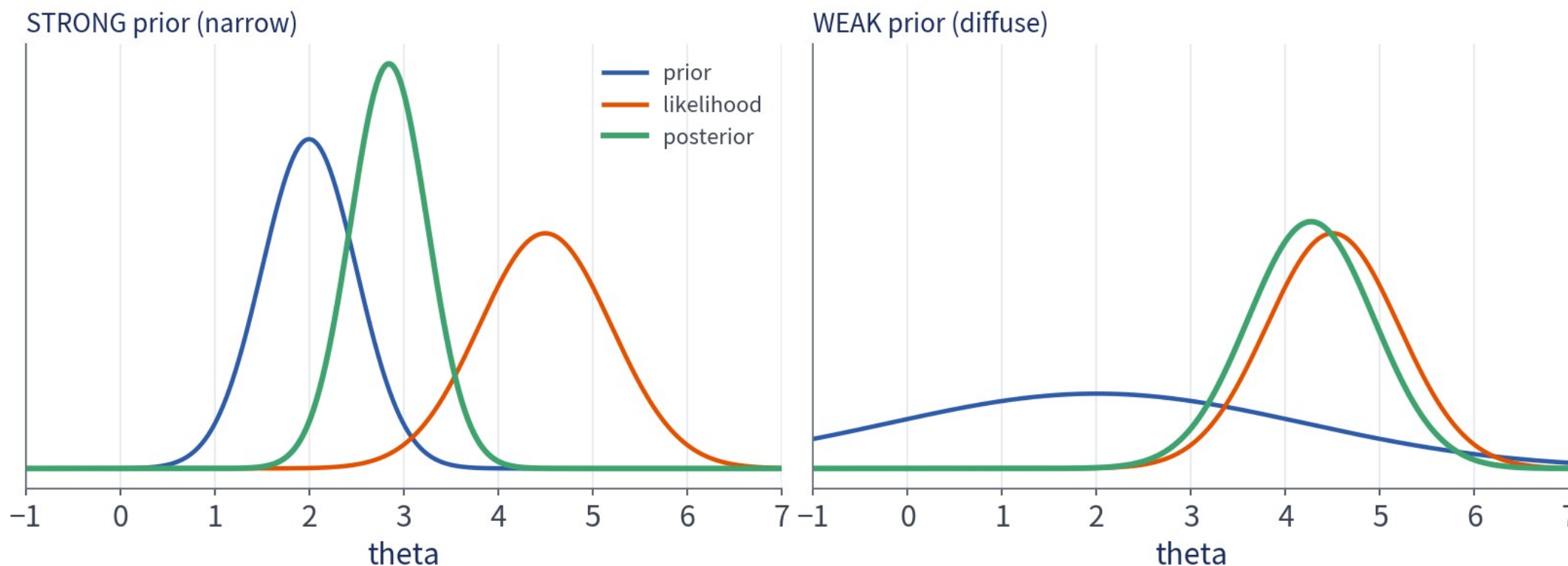
The healthy behavior

- With little data, the prior matters more.
- With lots of data, the prior washes out.
- Belief and evidence trade off sensibly.

Prior Strength Shapes the Posterior

With the same data, a strong (narrow) prior holds the posterior near the prior belief, while a weak (diffuse) prior lets the likelihood lead. The prior's confidence decides how much the data moves you.

A strong prior pulls the posterior toward belief; a weak one lets the data lead



Parameters as Distributions, Not Points

The change in mindset that unlocks everything

The defining move of Bayesian thinking: a parameter is not a single unknown number but a distribution. Instead of 'the value is 0.62', you say 'here is the whole range of plausible values, with their probabilities'.

The old way

- Report one best estimate of the parameter.
- Uncertainty is a separate afterthought.
- The number hides how sure you are.

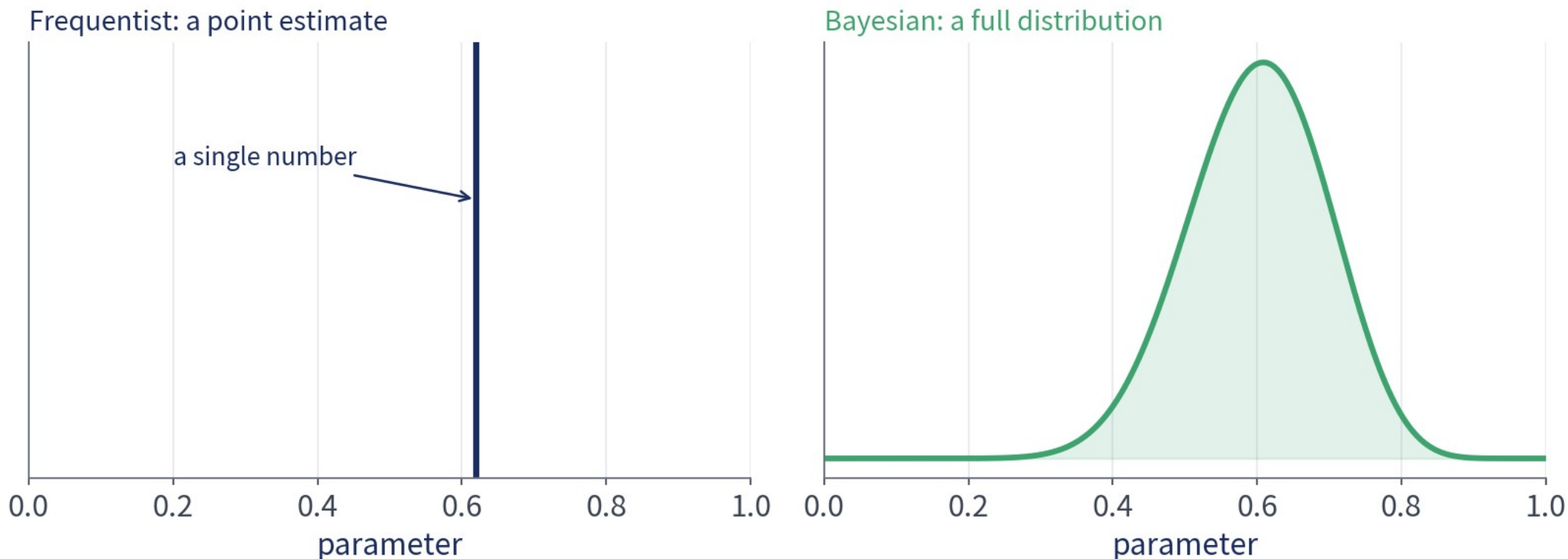
The Bayesian way

- Report a full distribution over the parameter.
- Its spread IS the uncertainty.
- Belief and confidence live in one object.

A Point Versus a Distribution

The frequentist reports a single number; the Bayesian reports a whole distribution. The width of that distribution is the honest measure of how much the data has, and hasn't, pinned the parameter down.

Parameters as distributions, not points



What a Posterior Tells You

More than a single number

It is a full belief

- The most probable value (its peak).
- The uncertainty around it (its spread).
- The probability of any range you care about.

What you can do with it

- Summarize it by a mean or a credible interval.
- Feed it forward into predictions.
- Combine it with more data later.

The Prior: Where Belief Starts

Your knowledge before the data

What it encodes

- What you believed before seeing this data.
- It can be informative or deliberately vague.
- It is a genuine input, chosen with care.

Why it is useful

- It injects real domain knowledge.
- It stabilizes estimates when data is scarce.
- It is also the most debated Bayesian ingredient.

The Likelihood: What the Data Says

The model linking parameter to observations

What it is

- The probability of the data for each parameter value.
- It is the same model a frequentist would use.
- It is where the observations enter.

Its role

- It reweights the prior toward data-consistent values.
- More data makes it sharper and more decisive.
- It is the evidence in Bayes' theorem.

The Evidence: The Normalizer

Why Bayes waited for computers

$$P(\text{data}) = \int P(\text{data} \mid \theta) P(\theta) d\theta \quad (\text{the normalizer - often the hard part})$$

What it is

- The overall probability of the data.
- It integrates the likelihood over all parameter values.
- It rescales the posterior into a proper distribution.

Why it is hard

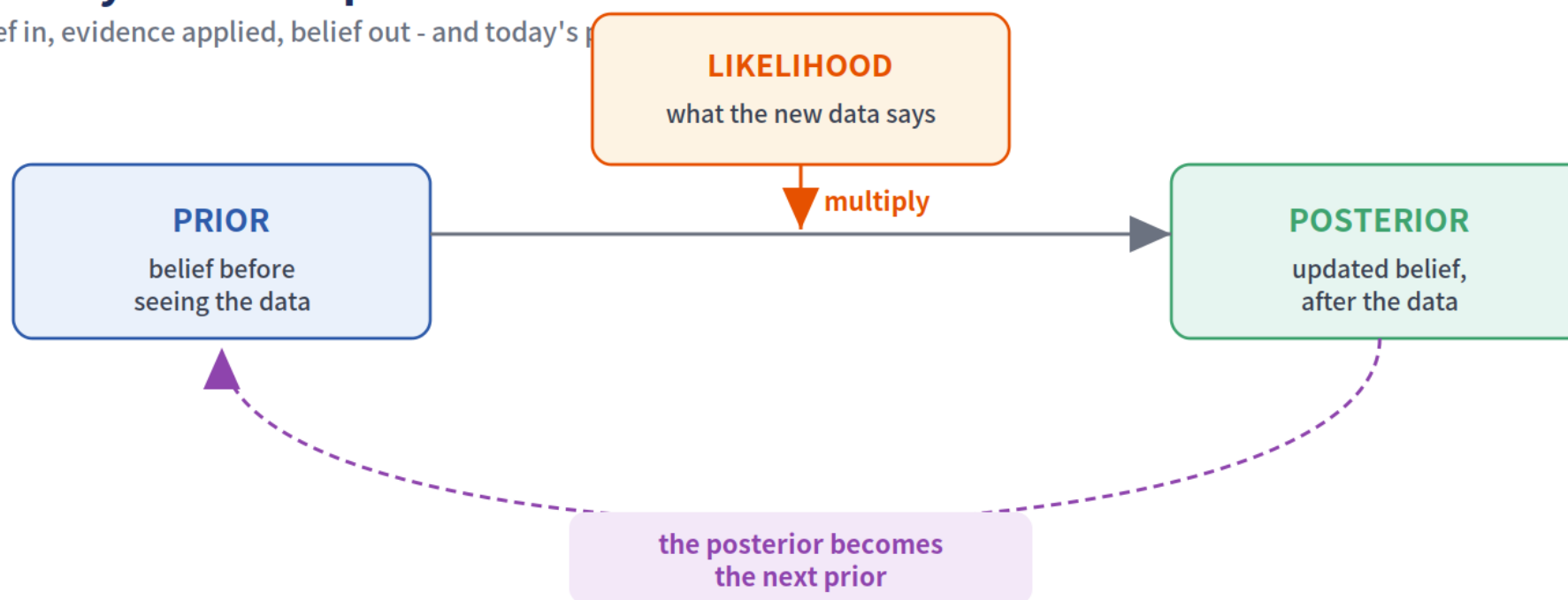
- That integral is usually intractable by hand.
- In many dimensions it is impossible to do exactly.
- This is the problem MCMC solves next week.

The Bayesian Loop

A prior meets new data through the likelihood and becomes a posterior. That posterior then serves as the prior for the next batch of data — learning is an iterative loop, not a one-shot calculation.

The Bayesian Loop

Belief in, evidence applied, belief out - and today's p



Learning is iterative: each batch of evidence updates the belief, and that belief carries forward to the next.

Bayes' Theorem by Hand

Direct Claude Code through one clean update

GOAL Do a single Bayesian update numerically and see the posterior emerge from prior times likelihood.

[INPUT] Prompt · Claude Code

Write and run `bayes_update.py` that:

1. puts a Normal PRIOR on a parameter `theta`;
2. defines a Normal LIKELIHOOD from some data;
3. forms the POSTERIOR on a grid as prior times likelihood, then normalizes it to integrate to 1;
4. plots prior, likelihood, and posterior together, and prints the posterior mean and a 95% interval.

Show the posterior lies between prior and likelihood and is narrower than either.

How to work with it

- Posterior = prior x likelihood, then normalize.
- It sits between the two inputs.
- It is tighter than either alone.
- The grid makes the normalizer explicit.

Deliverable: `bayes_update.py` + a prior-likelihood-posterior plot and a posterior summary.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 To a Bayesian, what is a parameter?

A distribution, not a fixed number

Q2 In the working form, the posterior is proportional to what?

Likelihood times prior

Q3 Which ingredient is the computationally hard part?

The evidence (normalizer)

All three come from the Bayesian mindset, the working form, and the normalizing integral.

Three Misconceptions to Avoid

Where Bayesian intuition slips

Common myths

- "The prior is just a fudge factor you make up."
- "With enough data, the prior still dominates."
- "The posterior is a single best estimate."

The reality

- The prior encodes real knowledge and can be tested.
- More data washes the prior out — the likelihood wins.
- The posterior is a full distribution, not one number.

Three Things to Carry Forward

Part 1 in one screen

1

Bayes updates belief with evidence

Bayes' theorem turns a prior belief and a likelihood into a posterior. In its working form, the posterior is proportional to likelihood times prior — belief reweighted by the data.

2

Parameters are distributions

Instead of one best number, a Bayesian carries a full distribution over the parameter. Its spread is the uncertainty, stated honestly rather than hidden behind a point estimate.

3

The normalizer is the catch

Computing the evidence — the integral that rescales the posterior — is the hard part. It is exactly why Bayesian methods waited for computers, and it motivates next week's MCMC.

Next — Part 2: the Beta-Binomial, sequential updating, conjugate priors, and the role of the prior.

★ WEEK 13 · PART 1 · PRACTICE

Bayes' Theorem, Worked by Hand

Update a Belief About Skill, and Blend a View With Data

The discrete update and the precision-weighted compromise

Part 1's update rule, with the arithmetic shown. We compute how a market-beating record shifts belief in a manager's skill - and why one good year proves little - then blend an analyst's prior view with observed returns by precision weighting.

1P

PART 1 · APPLICATIONS

Two Updates, by Hand

Case A: a discrete Bayes update on a manager's skill - one beat, then several, showing why base rates matter. Case B: the posterior as a precision-weighted compromise between a prior view and the data. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Bayes updates a belief with evidence

The posterior is proportional to the likelihood times the prior. New data reshapes what you believed before.

2

Parameters are distributions, not points

You don't get a single answer - you get a whole distribution of belief, with a center and a spread.

3

The update is a compromise

The posterior sits between the prior and the data, pulled toward whichever is more precise. We compute exactly where.

Everything here is a calculator exercise: one manager's record, one analyst's view and some data.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a prior and a likelihood.
- Apply Bayes to get the posterior by hand.
- Then read what the belief has become.

What you should watch

- How a low base rate tempers the evidence.
- How belief accumulates over observations.
- How precision decides the compromise.

Is This Manager Skilled?

One good year proves surprisingly little

Only a small fraction of managers are genuinely skilled, but plenty of unskilled ones beat the market by luck in any given year. So a market-beating record is evidence of skill - but weak evidence. We compute the posterior probability of skill after one beat, then after several, and watch belief accumulate

The ingredients

- Prior: 10% of managers are skilled.
- A skilled one beats the market 80% of years.
- An unskilled one beats it 30% by luck.

What we'll do by hand

- Compute the posterior after one beat.
- Feed it back as the prior for the next.
- Track belief over three beats.

Bayes' Theorem

Posterior from likelihood and prior

$$\underbrace{P(H \mid D)}_{\text{posterior}} = \frac{\overbrace{P(D \mid H)}^{\text{likelihood}} \overbrace{P(H)}^{\text{prior}}}{\underbrace{P(D)}_{\text{evidence}}} \propto \text{likelihood} \times \text{prior}$$

The posterior combines how well the hypothesis explains the data (the likelihood) with how plausible it was to begin with (the prior), normalized by the total evidence.

After One Market-Beating Year

The base rate holds belief down

$$P(\text{skilled} \mid \text{beat}) = \frac{0.10 \times 0.80}{0.10(0.80) + 0.90(0.30)} = \frac{0.08}{0.35} = \mathbf{0.23}$$

The calculation

- Skilled path: $0.10 \times 0.80 = 0.08$.
- Lucky path: $0.90 \times 0.30 = 0.27$.
- Posterior is 0.08 over their sum.

The surprise

- Even after beating the market, only 23%.
- The 10% base rate anchors the belief.
- One good year is weak evidence.

Belief Accumulates

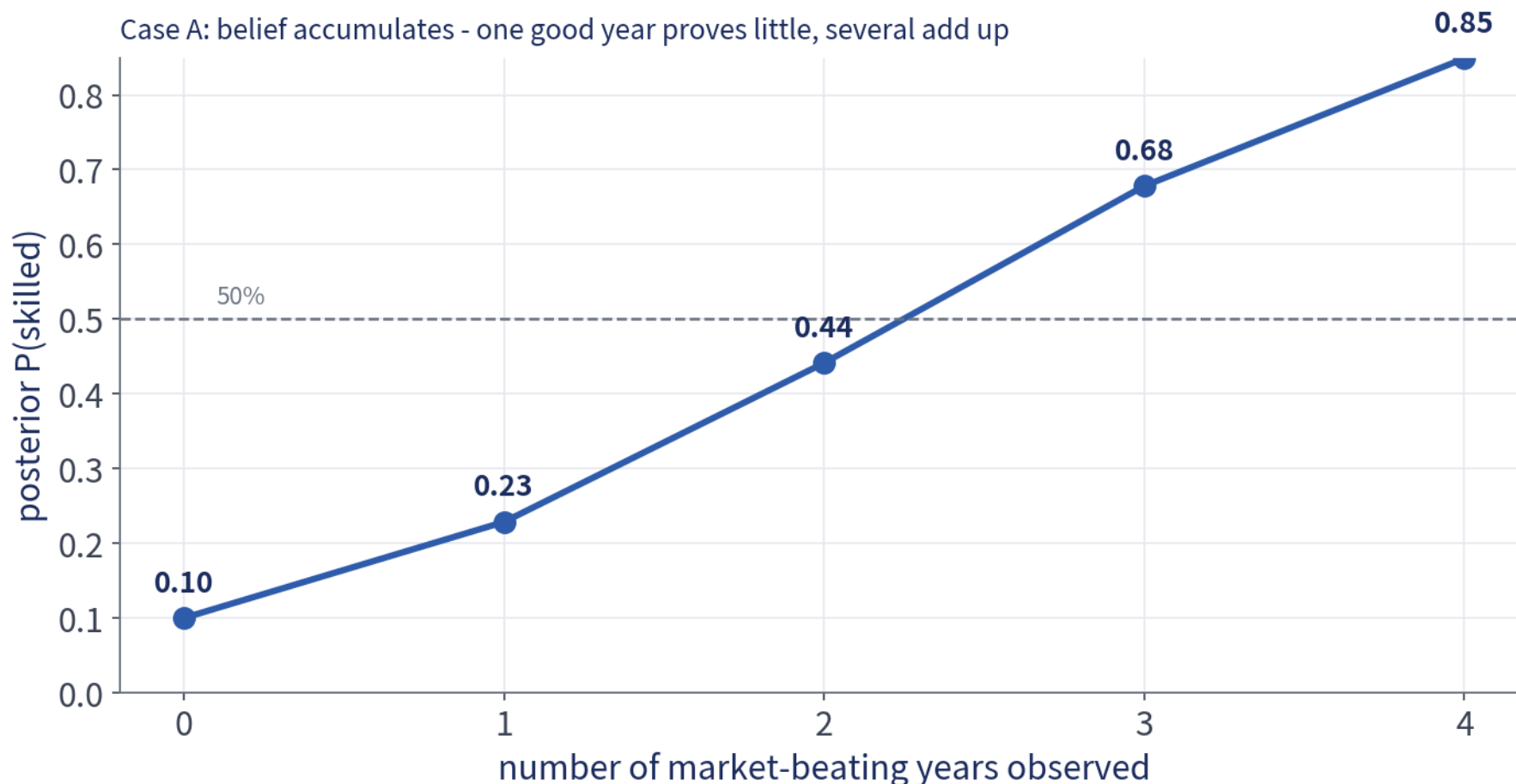
Each posterior becomes the next prior

1 beat: $0.10 \rightarrow \mathbf{0.23}$
2 beats: $0.23 \rightarrow \mathbf{0.44}$
3 beats: $0.44 \rightarrow \mathbf{0.68}$ (each posterior becomes the next prior)

Repeated beats compound the evidence: 23% after one year, 44% after two, 68% after three. This feedback - posterior becoming the next prior - is the Bayesian loop in action.

Belief Builds Year by Year

Starting from a 10% base rate, each market-beating year lifts the posterior - but slowly at first. It takes three consecutive good years to cross above even-odds. That's the discipline Bayes imposes: extraordinary claims need accumulated evidence



Why Base Rates Matter

The prior is not optional

The trap

- Seeing a good record and assuming skill.
- Ignoring how many beat the market by luck.
- Over-reacting to a single data point.

The discipline

- The prior keeps the evidence in proportion.
- Weak evidence moves belief only a little.
- Accumulated evidence is what convinces.

Blend a View With Data

The posterior as a compromise

An analyst expects an 8% return with fair confidence; the data shows a 14% average, but from a noisy sample. Bayes blends them into a posterior - a precision-weighted average, where each source counts in proportion to how sure it is. We compute the posterior mean and, crucially, its tighter spread

The two inputs

- Prior view: 8%, with a 3% spread.
- Data: 14% average, with a 4% standard error.
- The prior is the more precise of the two.

What we'll do by hand

- Turn each spread into a precision.
- Take the precision-weighted average.
- Combine precisions for the posterior spread.

The Precision-Weighted Mean

Each source counts by its certainty

$$\mu_{\text{post}} = \frac{\tau_0 \mu_0 + \tau_d \bar{x}}{\tau_0 + \tau_d}, \quad \tau = \frac{1}{\text{variance}} \text{ (precision)}$$

Precision is inverse variance

- A tight belief has high precision.
- A noisy one has low precision.
- It's how much a source should count.

The blend

- Weight prior and data by their precisions.
- The more certain source pulls harder.
- The result is the posterior mean.

The Posterior Mean

A compromise, nearer the tighter view

$$\mu_{\text{post}} = \frac{1111(0.08) + 625(0.14)}{1111 + 625} = \frac{176.4}{1736} = \mathbf{10.2\%} \quad (\text{a compromise, nearer the tighter prior})$$

The weights

- Prior precision 1111, data precision 625.
- So the prior counts about 64%.
- The data about 36%.

The result

- The posterior mean is 10.2%.
- Between the 8% view and 14% data,
- but pulled toward the tighter prior.

The Posterior Spread

Combining information sharpens belief

$$\tau_{\text{post}} = \tau_0 + \tau_d = 1736 \Rightarrow \sigma_{\text{post}} = \frac{1}{\sqrt{1736}} = \mathbf{2.4\%} \quad (\text{tighter than either input})$$

Precisions add

- Posterior precision is the sum: 1736.
- More information means more precision.
- So the spread shrinks.

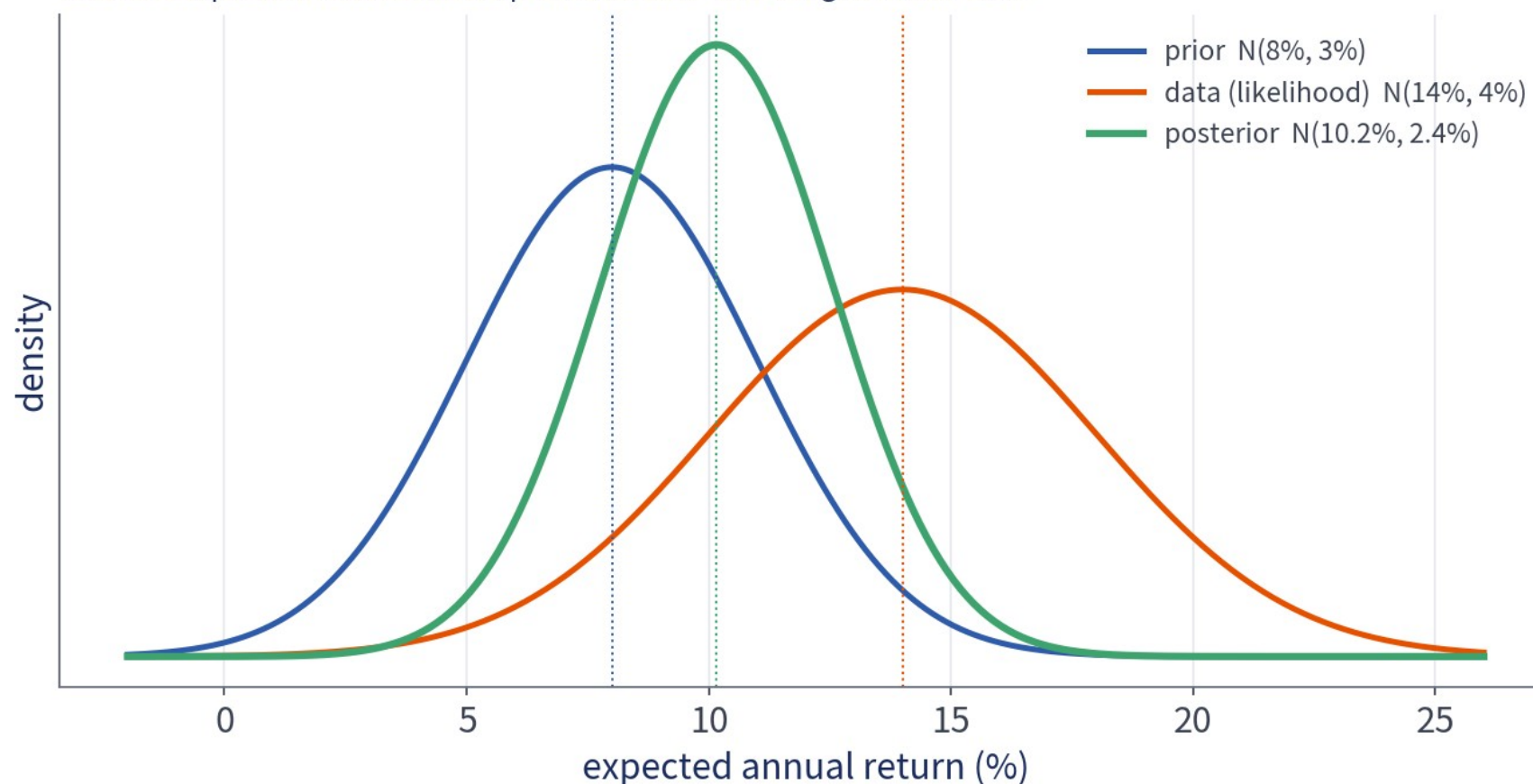
The payoff

- The posterior sd is 2.4%.
- Tighter than both the prior and the data.
- Two views together beat either alone.

Between, and Tighter

The posterior (green) sits between the prior (blue, at 8%) and the data (orange, at 14%), landing at 10.2% - closer to the tighter prior. And it's narrower than either: combining two independent views always sharpens the belief

Case B: the posterior sits between prior and data - and is tighter than both



The Update Is a Compromise

Precision decides who wins

What moves the belief

- A confident prior resists the data.
- Precise data overrides a vague prior.
- The posterior lands where precision points.

Why it matters in finance

- It blends a view with evidence, rigorously.
- It's the seed of Black-Litterman and shrinkage.
- And it always reduces your uncertainty.

The Bayesian Loop

Prior, evidence, posterior, repeat

Both cases are the same loop. A prior meets new data through the likelihood and becomes a posterior - which then serves as the prior for the next round. Whether it's a manager's record updating discrete odds or returns refining a continuous view, the machinery is identical: belief in, evidence, belief out

Discrete (Case A)

- Prior odds meet a likelihood ratio.
- Posterior odds emerge, then repeat.
- Belief accumulates over observations.

Continuous (Case B)

- A prior distribution meets the data.
- Precisions combine into a posterior.
- Tighter, and nearer the surer source.

Reproduce Both Updates

Direct Claude Code to check your arithmetic

GOAL Verify the discrete update and the precision-weighted blend, then vary the inputs.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - manager skill:

- * from prior 0.10, $P(\text{beat}|\text{skill})=0.80$,
 $P(\text{beat}|\text{luck})=0.30$, print the posterior after
1, 2, and 3 beats.

CASE B - precision blend:

- * from prior $N(0.08, 0.03)$ and data $N(0.14, 0.04)$,
print the posterior mean and sd.

Then plot the posterior probability vs the number of beats, and the three densities.

Match every printed number to the slides.

Check against the slides

- posteriors 0.23, 0.44, 0.68.
- prior counts ~64%.
- posterior mean 10.2%.
- posterior sd 2.4%.

Deliverable: `part1_practice.py` whose printed numbers match the discrete update and the blend.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why is one market-beating year weak evidence?

**A low base rate
anchors the posterior**

Q2 What is precision, and why weight by it?

**Inverse variance -
surer sources count
more**

Q3 Why is the posterior tighter than both inputs?

**Precisions add when
you combine
information**

All three come from the discrete update and the precision-weighted compromise.

Three Pitfalls to Avoid

Where Bayesian reasoning slips

Common mistakes

- "A good track record proves skill."
- "Ignore the prior and just follow the data."
- "Average the two views equally."

The reality

- With a low base rate, it only shifts belief a little.
- The prior keeps weak evidence in proportion.
- Weight by precision - the surer source counts more.

Two Updates to Remember

Part 1 applications in one screen

1

A discrete update tempers the evidence

With a 10% base rate, one market-beating year lifts belief in skill only to 23%; two years to 44%, three to 68%. Each posterior becomes the next prior - the loop - so belief accumulates rather than jumping on a single data point.

2

A continuous update is a precision-weighted blend

Blending a prior view of 8% (precision 1111) with data of 14% (precision 625) gives a posterior mean of 10.2% - nearer the tighter prior - and a posterior sd of 2.4%, tighter than either input. Precisions add.

3

One loop, two forms

Discrete odds or continuous distributions, the machinery is the same: prior meets data through the likelihood and becomes a posterior. That posterior is the prior for whatever comes next.

Next — Part 2 practice: the beta-binomial, where prior and posterior share a form and updating is trivial.

★ WEEK 13 · PART 2 of 3

Updating in Practice: The Beta-Binomial

Conjugate Priors and the Role of the Prior

The Beta-Binomial · sequential updating · conjugacy · choosing a prior

The abstract update becomes concrete: estimating a probability. The Beta-Binomial makes Bayes' theorem pure arithmetic, shows how belief sharpens with data, and lays bare the role of the prior.

2

PART 2 · UPDATING IN PRACTICE

The Beta-Binomial in Full

Estimating a probability with a Beta prior and Binomial data, the conjugate update as counting, how the posterior sharpens with data, and the much-debated role of the prior.

From the Rule to a Real Update

Part 1 gave the theory; Part 2 works an example

1

We have the update rule

Posterior is proportional to likelihood times prior. But abstractly it can feel slippery — we need a concrete case.

2

Estimating a probability is the classic case

How likely is an up-day? A default? The Beta-Binomial answers exactly this, and the update is pure arithmetic.

3

It reveals the key ideas

Conjugacy, sequential updating, and the role of the prior all show up cleanly in this one worked example.

Part 2 in one line: the Beta-Binomial turns Bayes' theorem into counting successes and failures.

Estimating a Probability

The Beta-Binomial: the 'hello world' of Bayes

The simplest useful Bayesian problem: estimate a probability from yes/no data. Put a Beta prior on the probability, observe Binomial data, and the posterior is again a Beta — the update is just arithmetic.

The question

- What is the probability of a success?
- Examples: an up-day, a default, a win.
- We want a distribution over that probability.

The ingredients

- A Beta prior on the probability.
- A Binomial likelihood for the counts.
- A Beta posterior — same family, updated.

The Binomial Likelihood

Counting successes out of trials

Binomial likelihood: $P(k | n, \theta) \propto \theta^k (1 - \theta)^{n-k}$

What it says

- Given a probability, how likely is k successes in n ?
- It rewards values near the observed proportion.
- It is the model linking the probability to the data.

Its shape

- Peaked at the sample proportion k over n .
- Sharper as the number of trials grows.
- This is the likelihood in Bayes' theorem.

The Beta Prior

A distribution on a probability in $[0,1]$

$$\text{Beta prior: } \theta \sim \text{Beta}(a, b), \quad p(\theta) \propto \theta^{a-1}(1 - \theta)^{b-1}$$

Why the Beta

- It lives exactly on the interval zero to one.
- So it is a natural prior for a probability.
- Its two parameters a and b flex its shape.

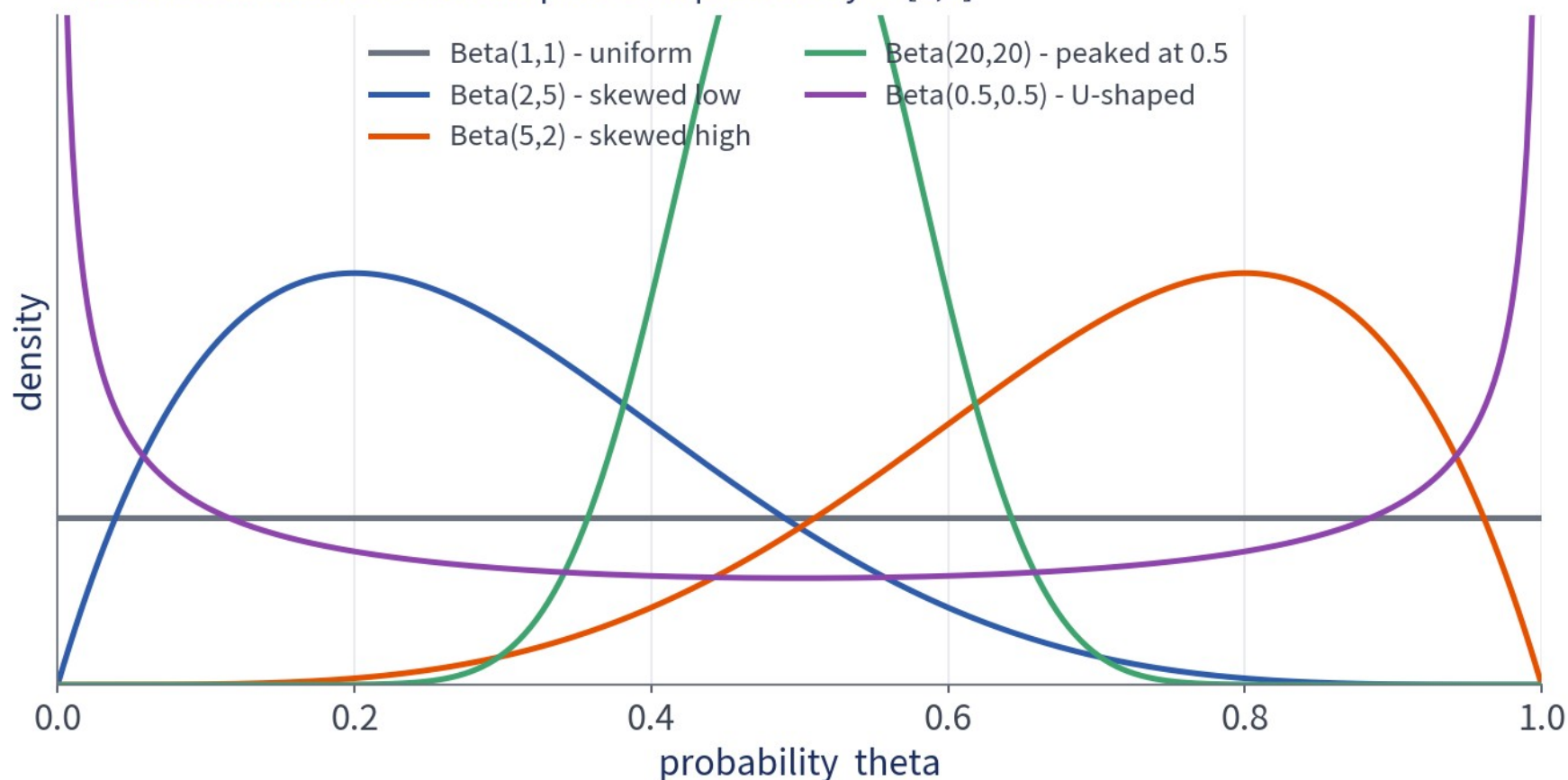
Reading a and b

- They behave like prior successes and failures.
- Larger values make a more confident prior.
- Equal values center it at one half.

The Beta Distribution's Many Shapes

By varying a and b , the Beta can be flat, bell-shaped, skewed, or U-shaped — a flexible family for expressing almost any prior belief about a probability between zero and one.

The Beta distribution: a flexible prior on a probability in $[0,1]$



The Beta-Binomial Update

Add the counts, get a new Beta

$$\text{Beta-Binomial update: } \theta \mid \text{data} \sim \text{Beta}(a + k, b + n - k)$$

The rule

- Add the observed successes k to the prior's a .
- Add the observed failures to the prior's b .
- That's the entire update — no integral needed.

Why it is remarkable

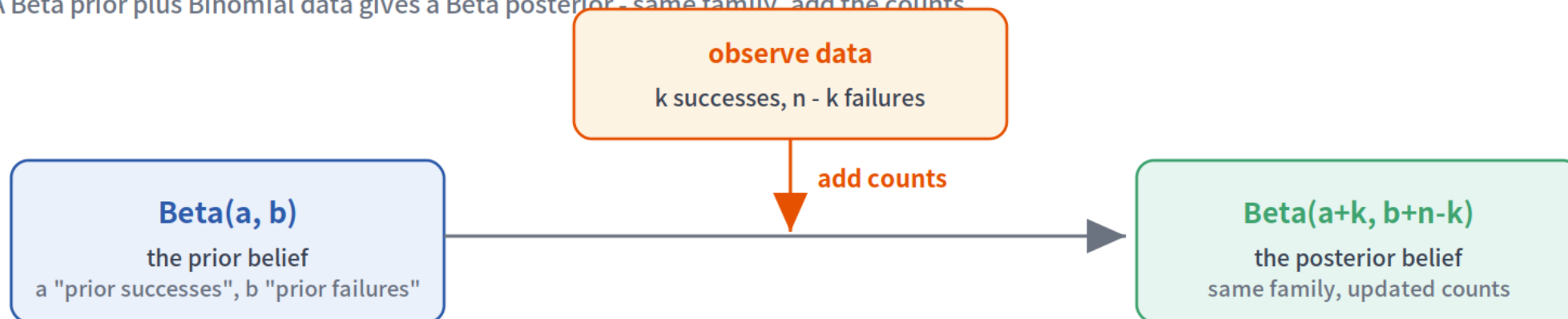
- The posterior is again a Beta distribution.
- So it can serve as the prior for the next batch.
- Bayes' theorem reduces to bookkeeping.

Conjugacy: the Update Is Just Counting

A Beta prior and Binomial data produce a Beta posterior — the same family, with the observed successes and failures simply added to a and b . No integral, no simulation: pure arithmetic.

Conjugacy: the Update Is Just Counting

A Beta prior plus Binomial data gives a Beta posterior — same family, add the counts



Why this is a gift

When the prior and likelihood are CONJUGATE, the posterior is in the same family as the prior. No integral, no simulation - just add the observed successes and failures to the prior's parameters.

a and b act like pseudo-counts: prior evidence, expressed as imaginary past data.

Walking Through the Update

From prior to posterior in one step

Set-up

- Start with a mild prior, say Beta two and two.
- It is centered at one half, weakly held.
- It reflects 'probably balanced, but unsure'.

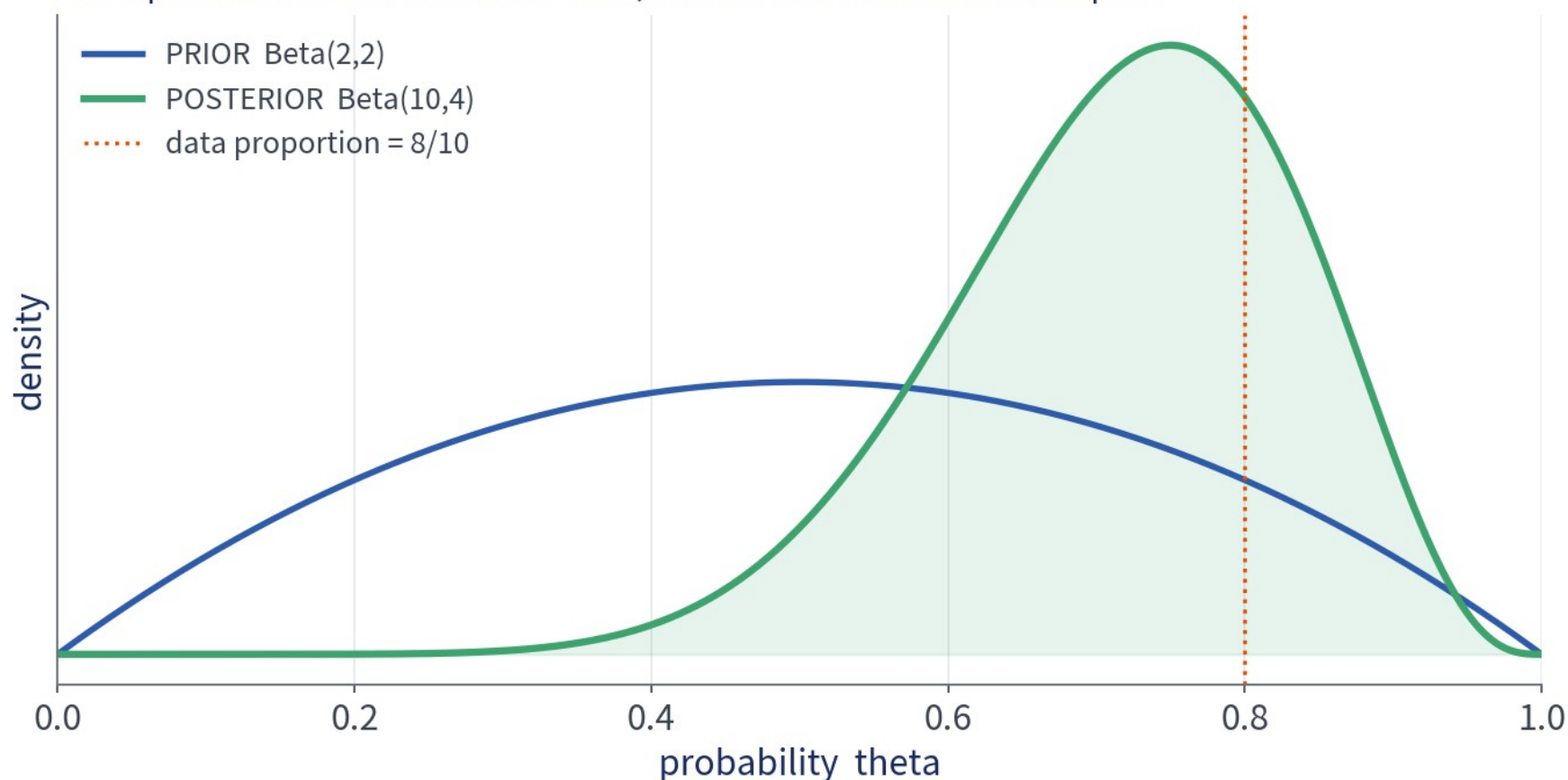
Observe and update

- See eight successes out of ten trials.
- Add the counts: the posterior is Beta ten and four.
- Belief moves toward 0.8 and tightens.

One Update, Start to Finish

A mild prior meets eight successes in ten trials. The posterior shifts toward the observed proportion and narrows — belief has moved decisively, but still carries honest uncertainty.

One update: observe 8 successes in 10, and the belief shifts and sharpens



The Posterior Sharpens With Data

Sequential updating, one Beta at a time

Because the posterior is again a Beta, you can keep updating. Each new observation adds to the counts, and the distribution tightens around the truth. Yesterday's posterior is simply today's prior.

How it accumulates

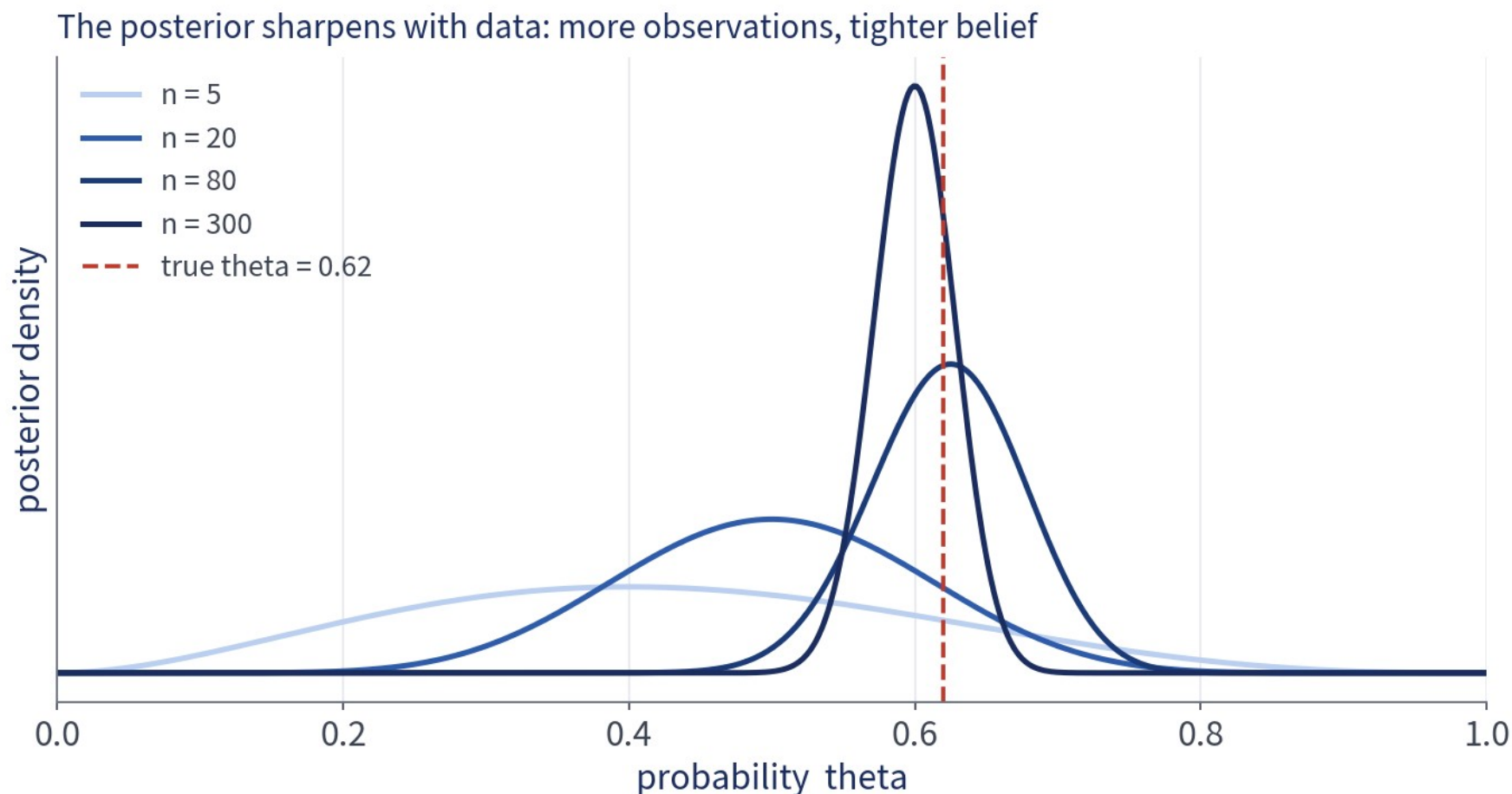
- Feed observations one at a time, or in batches.
- Each just increments a or b .
- The order does not change the final result.

What happens to belief

- The posterior narrows as evidence mounts.
- It concentrates around the true probability.
- Uncertainty shrinks in a principled way.

The Posterior Sharpens With Data

Starting from a flat prior, the posterior tightens around the true probability as observations accumulate. With a handful of points it is broad; with hundreds it is a confident spike.



Conjugate Priors

When the update is just arithmetic

A prior is conjugate to a likelihood when the posterior stays in the same family. Then Bayes' theorem needs no integral — you just update the parameters. The Beta-Binomial is the classic example.

What conjugacy means

- Prior and posterior share the same family.
- The likelihood maps one member to another.
- The update is a parameter change, nothing more.

Why it matters

- No intractable normalizing integral to compute.
- Fast, exact, and easy to reason about.
- It made Bayes usable long before computers.

Common Conjugate Pairs

A small toolbox of exact updates

Prior	Likelihood	Posterior
Beta	Binomial	Beta
Gamma	Poisson	Gamma
Normal (known variance)	Normal	Normal
Dirichlet	Multinomial	Dirichlet

Each pair updates by simple arithmetic — the reason conjugate models are so widely used.

The Posterior Mean as Shrinkage

A blend of prior and data

$$\text{posterior mean} = \frac{a+k}{a+b+n} \quad (\text{a shrinkage of the data toward the prior})$$

Reading it

- The mean blends prior counts with observed counts.
- It sits between the prior mean and the data proportion.
- It shrinks the raw estimate toward the prior.

Why shrinkage helps

- With little data, it leans on the prior — more stable.
- With much data, the observed counts dominate.
- It avoids wild estimates from tiny samples.

The Update as Counting

Priors as pseudo-counts

Prior as imaginary data

- The prior's a and b act like past successes and failures.
- A Beta two and two is like having seen two of each.
- The prior is 'pretend evidence' you start with.

Data as more counts

- Real observations add to those pseudo-counts.
- The posterior mean is just total successes over total trials.
- Big priors resist data; small priors yield to it.

The Role of the Prior

The most debated — and most useful — ingredient

The prior is where Bayesians are challenged: isn't it subjective? Yes — but it is an honest, explicit input, and a powerful one. It injects knowledge, stabilizes small samples, and fades as data accumulates.

The criticism

- The prior is a choice, and choices can be arguable.
- Different analysts might pick different priors.
- So conclusions can depend on that choice.

The defense

- The prior is explicit and open to scrutiny.
- It encodes genuine domain knowledge.
- With enough data, its influence disappears.

Informative vs Weak Priors

How much to let belief speak

Informative prior

- Encodes real, specific prior knowledge.
- Narrow — it resists being moved by little data.
- Powerful when you truly know something.

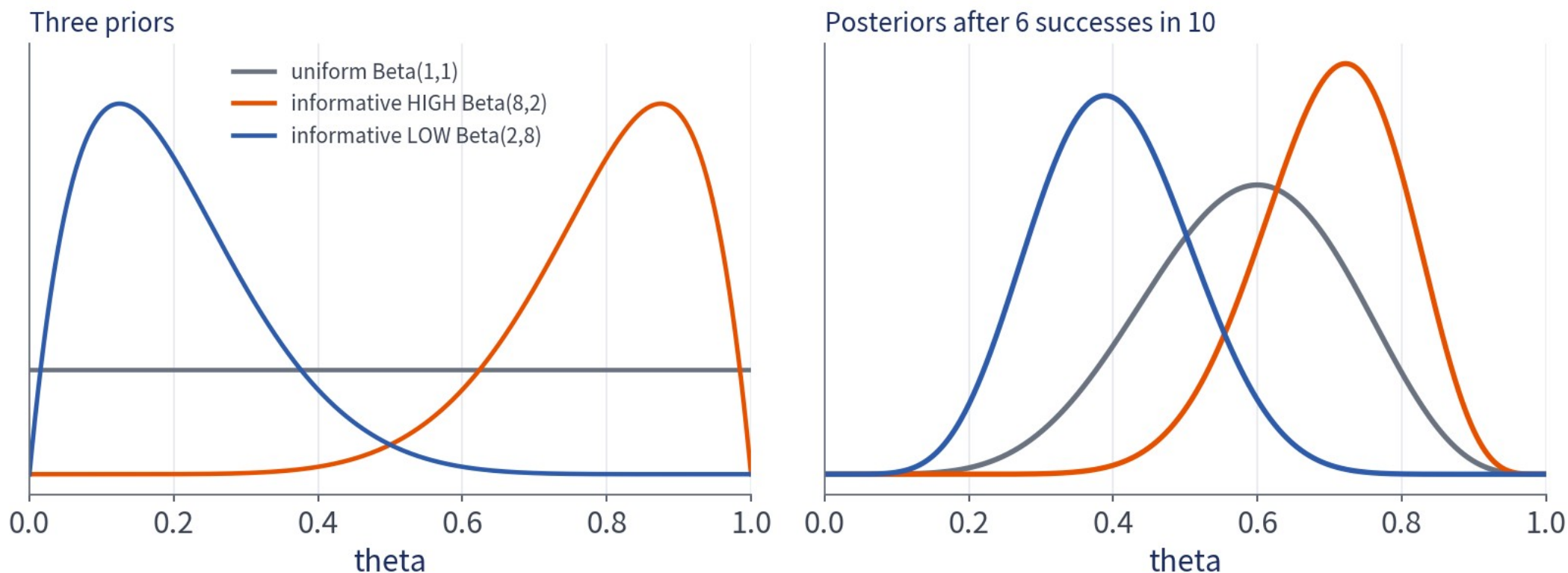
Weak (diffuse) prior

- Deliberately vague — lets the data lead.
- Wide and nearly flat over the range.
- A safe default when you know little.

Same Data, Different Priors

With only ten observations, three different priors yield three different posteriors. The prior still matters when data is limited — which is exactly when its stabilizing knowledge is most valuable.

Same data, different priors: the prior still tilts the posterior when data is limited



Priors as Regularization

Stabilizing small-sample estimates

The problem it solves

- Tiny samples give wild, unstable estimates.
- Zero successes doesn't mean zero probability.
- Raw proportions can be nonsensical at the extremes.

How the prior helps

- It pulls estimates toward a sensible default.
- This shrinkage is a form of regularization.
- It trades a little bias for much less variance.

When the Prior Washes Out

And how to choose one responsibly

The washout

- As data grows, the likelihood dominates.
- The posterior converges regardless of a reasonable prior.
- Priors matter most exactly when data is scarce.

Choosing responsibly

- Prefer weakly-informative priors as a default.
- Run a sensitivity check across plausible priors.
- Report how much the prior actually mattered.

Beta-Binomial Sequential Updating

Direct Claude Code to watch belief sharpen

GOAL Estimate a probability sequentially with a Beta-Binomial and see the posterior tighten as data arrives.

[INPUT] Prompt · Claude Code

Write and run `beta_binomial.py` that:

1. sets a true probability and simulates a stream of Bernoulli (yes/no) outcomes;
2. starts from a $\text{Beta}(a, b)$ prior and updates it observation by observation (add to a or b);
3. plots the posterior after 5, 20, 80, and 300 observations, with the true value marked;
4. also compares two different priors on the same data to show the prior washing out.

Confirm the posterior concentrates on the truth.

How to work with it

- A success adds one to a ; a failure to b .
- The posterior narrows as n grows.
- Different priors converge with enough data.
- The posterior mean is a shrinkage estimate.

Deliverable: `beta_binomial.py` + sequential posteriors and a prior-sensitivity comparison.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

After k successes and $n-k$ failures, a $\text{Beta}(a,b)$ prior becomes what?

$\text{Beta}(a+k, b+n-k)$

Q2

What does 'conjugate' mean for a prior?

Posterior stays in the same family

Q3

When does the choice of prior matter most?

When data is scarce

All three come from the update rule, conjugacy, and the prior's fading influence.

Three Misconceptions to Avoid

Where Beta-Binomial intuition slips

Common myths

- "A conjugate prior is chosen because it's true."
- "The prior always biases the answer."
- "The order of the data changes the posterior."

The reality

- Conjugacy is chosen for convenience, not truth.
- With enough data the prior washes out entirely.
- For this model, the posterior is order-independent.

Three Things to Carry Forward

Part 2 in one screen

1

The Beta-Binomial is Bayes made concrete

A Beta prior and Binomial data give a Beta posterior. The update is pure arithmetic: add the observed successes and failures to the prior's parameters.

2

Conjugacy makes updating trivial

When prior and likelihood are conjugate, the posterior stays in the same family — no integral needed. The prior's parameters act as pseudo-counts, and the posterior mean is a shrinkage estimate.

3

The prior is honest and fades

It injects knowledge and stabilizes small samples, and it matters most when data is scarce. As evidence mounts, the likelihood takes over and the prior washes out.

Next — Part 3: credible intervals, the posterior predictive, and why Bayes suits finance.

★ WEEK 13 · PART 2 · PRACTICE

The Beta-Binomial, Worked by Hand

Estimate a Win Rate, and Tune the Prior's Strength

Conjugacy makes the update just counting

Part 2's conjugate update, with the arithmetic shown. We estimate a strategy's win rate by adding counts to a Beta prior - watching the estimate shrink from the raw frequency - then see how the prior's strength and the sample size shape the posterior.

2P

PART 2 · APPLICATIONS

Update by Counting

Case A: estimate a trading strategy's win rate with a Beta-Binomial update, and compare the Bayesian estimate to the naive frequency. Case B: priors as pseudo-counts - weak versus strong - and how the posterior sharpens with data. Then a Claude Code lab.

What Part 2 Told Us

Now we count - with numbers

1

A Beta prior lives on a probability

It's a flexible distribution on the interval from zero to one - perfect for a win rate or a default probability.

2

Conjugacy makes the update arithmetic

A Beta prior plus Binomial data gives a Beta posterior - you just add the wins and losses to the parameters.

3

The prior is pseudo-counts

A Beta's parameters act like trades you've already seen. Their size is how strongly the prior speaks against the data.

Everything here is a calculator exercise: one strategy's record, two priors, two sample sizes.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a Beta prior and some counts.
- Add the counts to get the posterior.
- Then read the mean and the spread.

What you should watch

- How the posterior shrinks toward the prior.
- How prior strength changes the pull.
- How more data sharpens the belief.

Estimate a Strategy's Win Rate

The raw frequency isn't the whole story

A strategy has won 12 of its last 20 trades - a 60% raw win rate. But 20 trades is a thin record. Starting from a mildly skeptical prior, the Beta-Binomial update just adds the wins and losses to the prior's counts, giving a posterior that tempers the raw frequency. We compute it by hand

The inputs

- Prior: Beta(4, 6), a mean of 0.40.
- Data: 12 wins and 8 losses.
- Naive win rate: 12 over 20, or 0.60.

What we'll do by hand

- Add the counts to the prior.
- Compute the posterior mean.
- Compare it to the raw frequency.

The Beta-Binomial Update

Conjugacy: just add the counts

$$\text{Beta}(\alpha, \beta) + \text{data } (s \text{ wins, } f \text{ losses}) \longrightarrow \text{Beta}(\alpha + s, \beta + f) \quad (\text{just add the counts})$$

Why it's so clean

- The Beta prior and Binomial data match.
- The posterior is Beta again - same family.
- No integration, just addition.

The recipe

- Add wins to the first parameter.
- Add losses to the second.
- That's the entire update.

The Posterior

Add the counts, read the mean

$$\begin{aligned}\text{posterior} &= \text{Beta}(4 + 12, 6 + 8) = \text{Beta}(16, 14) \\ \text{mean} &= \frac{16}{16 + 14} = \frac{16}{30} = \mathbf{0.533}\end{aligned}$$

Adding 12 wins and 8 losses to Beta(4,6) gives Beta(16,14). Its mean - the first parameter over their sum - is 0.533, the Bayesian estimate of the win rate.

Bayesian vs Naive

The estimate is pulled toward the prior

$$\underbrace{0.60}_{\text{naive 12/20}} \longrightarrow \underbrace{0.533}_{\text{Bayesian}} \longleftarrow \underbrace{0.40}_{\text{prior}} \quad (\text{shrunk toward the prior on thin data})$$

The gap

- The raw frequency says 60%.
- The Bayesian estimate says 53%.
- The prior of 40% pulled it down.

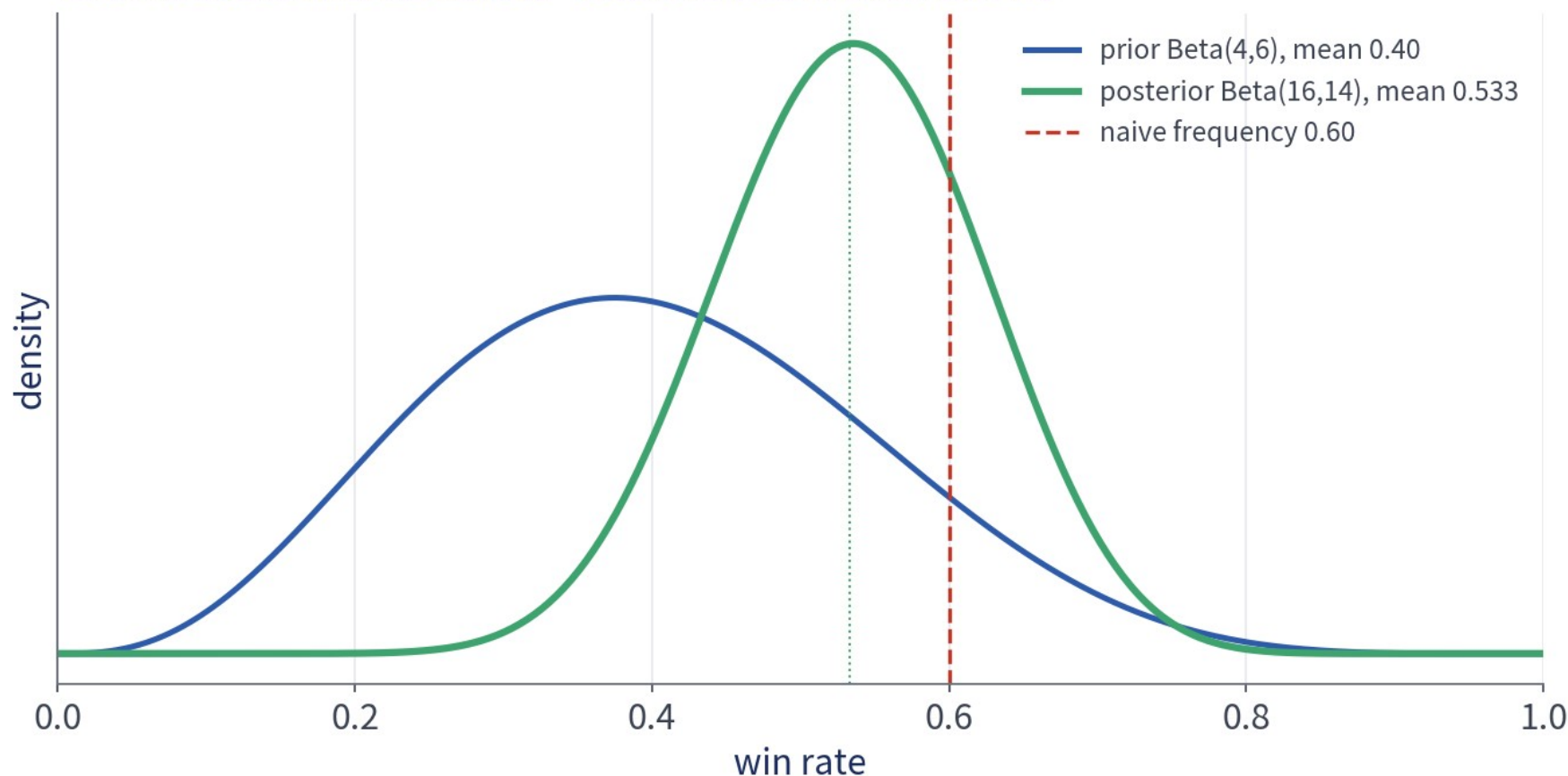
Why it's better

- Twenty trades is a small sample.
- The prior guards against over-reading it.
- The estimate is more robust to noise.

The Update Shifts and Sharpens

The broad prior (blue, centered at 0.40) becomes a tighter posterior (green, at 0.533) once the data arrives. It moves toward the observed frequency of 0.60 but stops short - the prior holds it back, exactly as a thin sample warrants

Case A: the update shifts and sharpens - posterior lands short of the naive 0.60



Shrinkage Is Protection

The prior tames a noisy estimate

The trap

- Taking a 60% win rate at face value.
- Sizing a strategy on 20 lucky trades.
- Chasing a number that won't hold up.

The discipline

- The prior shrinks the estimate toward reality.
- Small samples move belief only so far.
- You get a full distribution, not one number.

How Strong Is the Prior?

Pseudo-counts, and sharpening

A Beta prior's parameters are pseudo-counts: Beta(20,30) is like having already seen 50 trades. So a strong prior resists new data, while a weak one lets the data speak. And as real trades accumulate, the posterior sharpens regardless. We compute both effects by hand

Prior strength

- Weak Beta(1,1): two pseudo-trades.
- Strong Beta(20,30): fifty pseudo-trades.
- Same data, 12 wins and 8 losses.

Sharpening

- Compare 20 trades to 200 trades.
- Same win proportion, more evidence.
- Watch the posterior spread shrink.

Priors as Pseudo-Counts

Prior strength is injected data

$\text{Beta}(\alpha, \beta) \equiv \alpha + \beta$ pseudo-trades seen in advance (prior strength = how much data you inject)

The interpretation

- The parameters are like prior observations.
- Their sum is the prior's effective sample size.
- Bigger sum, stronger belief.

The consequence

- A strong prior needs a lot of data to move.
- A weak prior yields to even a small sample.
- You choose how loudly the prior speaks.

Weak vs Strong Prior

Same data, different pull

weak $\text{Beta}(1, 1) + 12W/8L = \text{Beta}(13, 9)$, mean **0.59** (follows data)
strong $\text{Beta}(20, 30) + 12W/8L = \text{Beta}(32, 38)$, mean **0.46** (resists)

The same 12-8 record gives a posterior mean of 0.59 under the weak prior - close to the data - but only 0.46 under the strong prior, which holds firm near its 0.40 belief.

The Posterior Sharpens

Ten times the data, half the spread

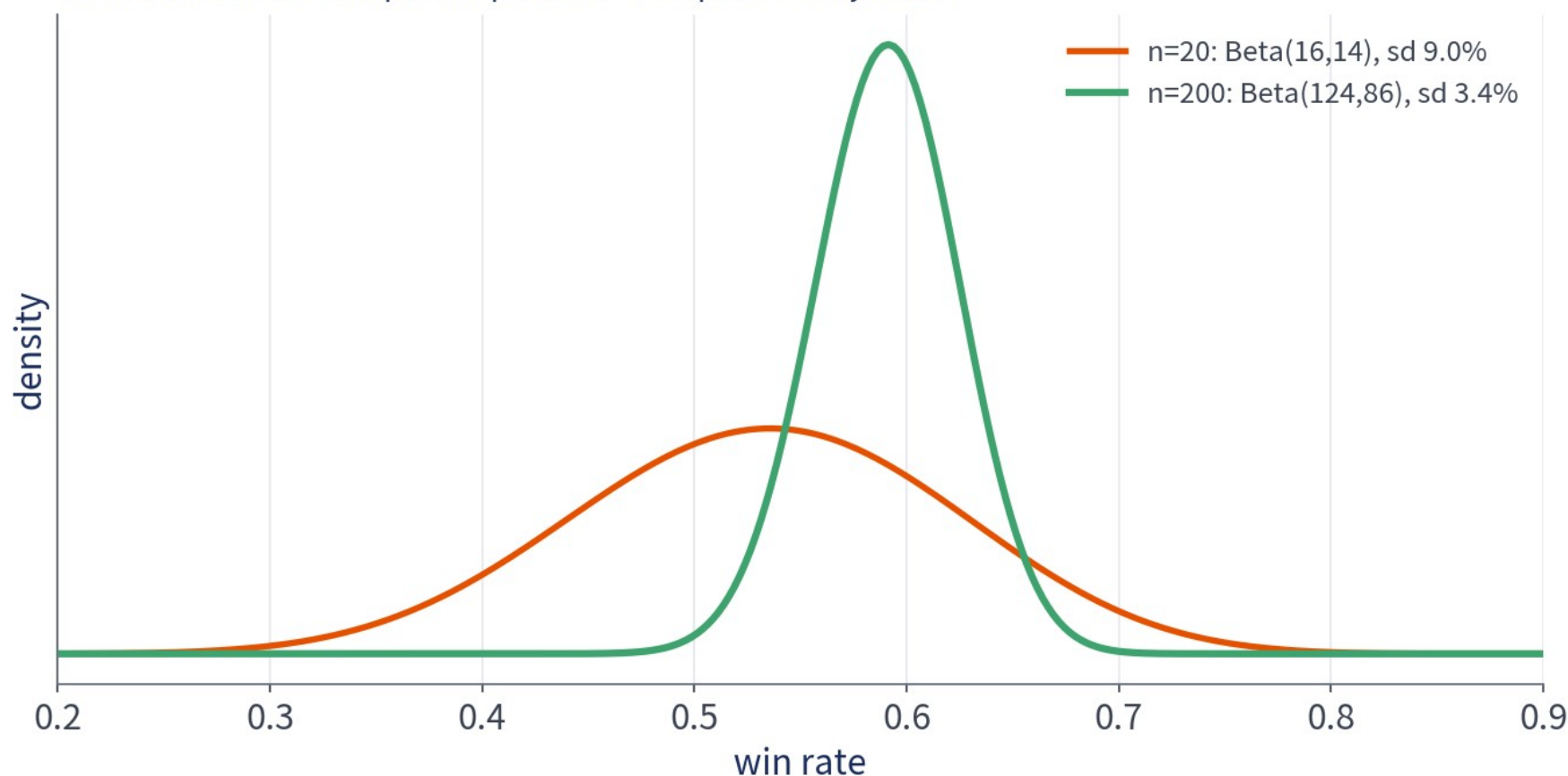
$$\begin{aligned} n = 20 & : \text{Beta}(16, 14), \text{ sd} = 9.0\% \\ n = 200 & : \text{Beta}(124, 86), \text{ sd} = 3.4\% \quad (\text{more data} \Rightarrow \text{sharper posterior}) \end{aligned}$$

With the same win proportion, going from 20 to 200 trades tightens the posterior from a 9.0% spread to 3.4%. More evidence always sharpens the belief.

More Data, Sharper Belief

With 20 trades the posterior (orange) is broad - the win rate could plausibly be anywhere from 0.35 to 0.70. With 200 trades at the same proportion (green), it collapses to a narrow peak. The spread nearly halves; the belief becomes decisive

Case B: more trades sharpen the posterior - the spread nearly halves



Choose the Prior Deliberately

Its strength is a decision

When to lean on the prior

- Little data, or a strong theoretical anchor.
- A strong prior stabilizes the estimate.
- But don't let it drown genuine signal.

When to let data speak

- Plenty of trades, or a weak prior belief.
- The data will dominate anyway.
- The posterior converges on the truth.

Conjugacy Is a Shortcut

When updating is arithmetic

The Beta-Binomial is the cleanest case of a general idea: conjugate priors, where prior and posterior share a family and the update is pure arithmetic. Add wins and losses to a Beta; add observations to a Normal; add counts to a Gamma. The prior's parameters are always pseudo-data, and more real data always sharpens the belief

The update (Case A)

- Beta prior plus counts, Beta posterior.
- Mean shrinks toward the prior.
- Robust on thin data.

The strength (Case B)

- Parameters are pseudo-counts.
- Strong resists, weak yields.
- More data sharpens either way.

Reproduce the Update and the Sharpening

Direct Claude Code to check your arithmetic

GOAL Verify the Beta-Binomial update, the prior-strength effect, and the sharpening.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - the update:

- * from `Beta(4,6)` and 12 wins / 8 losses, print the posterior Beta and its mean; compare to 12/20.

CASE B - strength and sharpening:

- * update `Beta(1,1)` and `Beta(20,30)` with the same data; print both posterior means;
- * print the posterior sd for `n=20` and `n=200`.

Then plot the prior and posterior densities, and the posterior for the two sample sizes.

Match every printed number to the slides.

Check against the slides

- posterior `Beta(16,14)`, mean 0.533.
- naive 0.60, prior 0.40.
- weak 0.59, strong 0.46.
- sd 9.0% -> 3.4%.

Deliverable: `part2_practice.py` whose printed numbers match the update, the strengths, and the sharpening.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you update a Beta prior with Binomial data?

**Add wins and losses
to its two parameters**

Q2 What does a prior's strength represent?

**Pseudo-counts -
trades seen in
advance**

Q3 What happens to the posterior as data grows?

**It sharpens - the
spread shrinks**

All three come from the Beta-Binomial update, pseudo-counts, and the sharpening.

Three Pitfalls to Avoid

Where the Beta-Binomial trips people

Common mistakes

- "The raw win rate is the win rate."
- "A strong prior is always safer."
- "A wide posterior means the model failed."

The reality

- On thin data, shrink it toward a sensible prior.
- Too strong a prior can drown a real signal.
- A wide posterior just means you need more data.

Two Calculations to Remember

Part 2 applications in one screen

1

Update a win rate by counting

Adding a 12-8 record to a $\text{Beta}(4,6)$ prior gives $\text{Beta}(16,14)$ with a mean of 0.533 - shrunk from the naive 0.60 toward the prior's 0.40, because 20 trades is a thin sample. Conjugacy made the update pure arithmetic.

2

The prior's strength is pseudo-counts

The same data gave a posterior mean of 0.59 under a weak $\text{Beta}(1,1)$ but only 0.46 under a strong $\text{Beta}(20,30)$ - fifty pseudo-trades that resist the evidence. And going from 20 to 200 real trades tightened the spread from 9.0% to 3.4%.

3

Conjugacy is the shortcut

Prior and posterior share a family, so updating is just adding counts. The parameters are pseudo-data, prior strength is a deliberate choice, and more real data always sharpens the belief.

Next — Part 3 practice: credible intervals, prediction, and putting Bayesian estimates to work in finance.

★ WEEK 13 · PART 3 of 3

Intervals, Prediction, and Why Bayes Suits Finance

Credible Intervals, the Posterior Predictive, and the Catch

Credible vs confidence · posterior predictive · the computing problem

A posterior does more than name a value: it gives credible intervals and honest forecasts. We contrast it with the frequentist interval, show why finance loves it, and meet the catch that MCMC will solve.

3

PART 3 · INTERVALS & PREDICTION

From Posterior to Decisions

Credible versus confidence intervals, the posterior predictive for honest forecasts, why Bayesian methods fit finance, and the computational catch that motivates Markov chain Monte Carlo.

From a Posterior to Answers

Parts 1-2 built the posterior; Part 3 uses it

1

We can compute a posterior

A full distribution over the parameter, from prior and data. Now: what do we actually DO with it?

2

It answers real questions

Credible intervals for uncertainty, and the posterior predictive for forecasts — both fall straight out of the posterior.

3

And it fits finance — with a catch

Short, noisy data make honest error bars essential. But computing posteriors for real models is hard, which sets up next week.

Part 3 in one line: read intervals and forecasts off the posterior — then confront the computing problem.

Credible vs Confidence Intervals

They are not the same thing

A credible interval and a confidence interval sound alike but mean different things. The credible interval is a probability statement about the PARAMETER; the confidence interval is a statement about the PROCEDURE across repeated samples.

Credible (Bayesian)

- 'There is a 95% probability θ is in this range.'
- A direct statement about the parameter.
- Read straight off the posterior.

Confidence (frequentist)

- '95% of such intervals would contain θ .'
- A statement about the method, not this θ .
- The parameter is fixed; the interval is random.

The Credible Interval

A direct probability about the parameter

$$\text{credible interval: } P(L \leq \theta \leq U \mid \text{data}) = 0.95$$

What it says

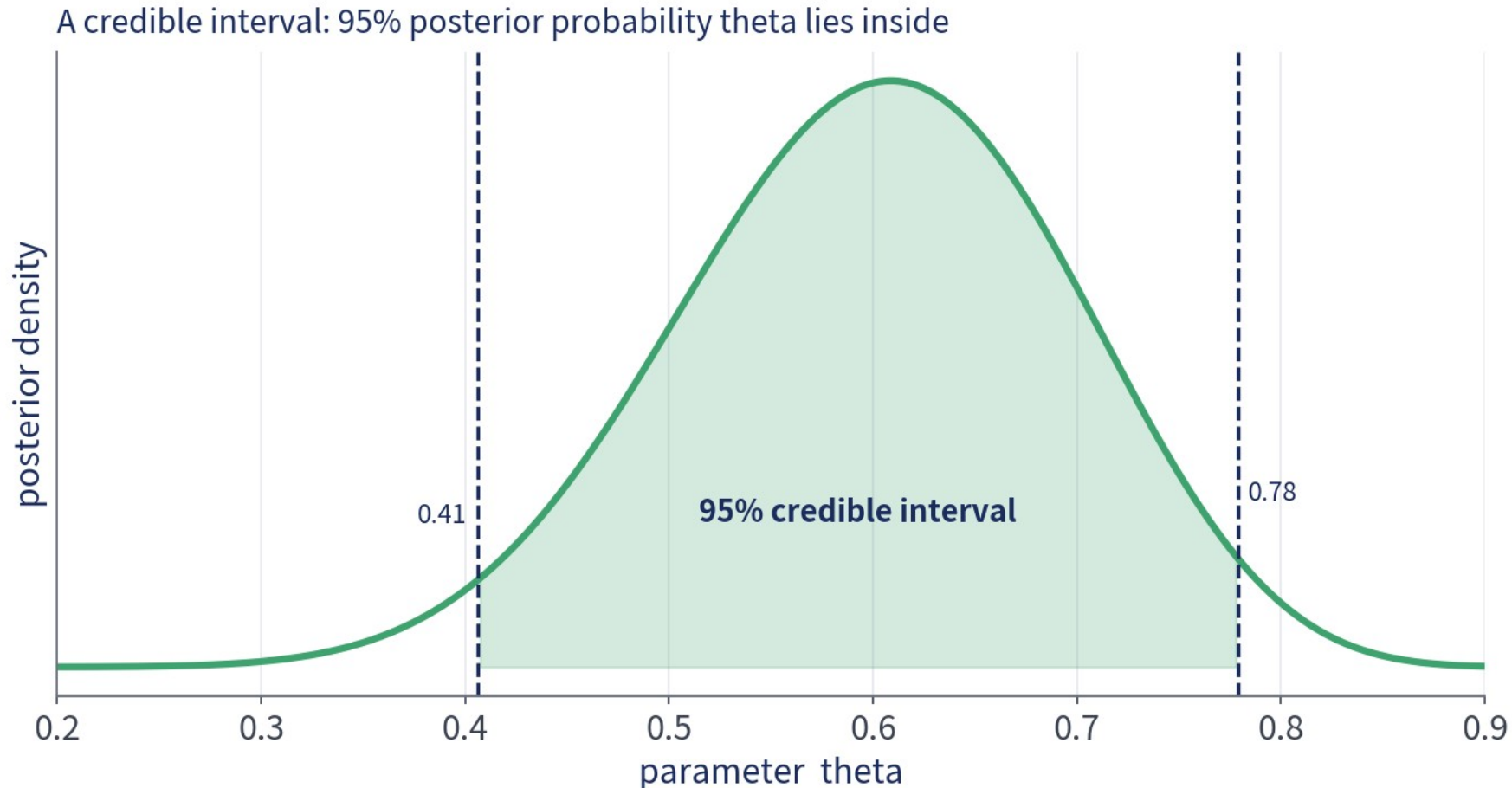
- The parameter lies in the range with 95% probability.
- That probability is given THIS data.
- It is exactly the intuitive reading people want.

How you get it

- Take the region holding 95% of the posterior.
- Equal-tailed, or the highest-density region.
- No repeated-sampling story needed.

A Credible Interval

Shade the central 95% of the posterior and you have the credible interval. It means exactly what it sounds like: 95% posterior probability that the parameter lies inside the shaded band.



The Confidence Interval

A statement about the procedure

What it actually means

- Imagine repeating the study many times.
- 95% of the intervals produced would cover θ .
- It is a property of the method, over repetitions.

What it does NOT mean

- It is NOT a 95% probability for THIS interval.
- The parameter is fixed; it is in or out.
- The common intuitive reading is technically wrong.

Two Kinds of Interval

The credible interval makes probability a statement about the parameter, given this data. The confidence interval makes it a statement about the procedure, over hypothetical repeated samples.

Two Kinds of Interval

They sound alike but say very different things - one about the parameter, one about the procedure.

Credible interval (Bayesian)

"Given THIS data, there is a 95% probability that θ lies in $[L, U]$."

- A statement about the PARAMETER.
- Uses this one dataset.
- Read straight off the posterior.
- Says what people usually MEAN.

Probability is about θ .

Confidence interval (frequentist)

"95% of such intervals, over many repeated samples, contain θ ."

- A statement about the PROCEDURE.
- Imagines many repeated samples.
- θ is fixed; the interval is random.
- NOT a probability about this θ .

Probability is about the method.

Why the Difference Matters

The interpretation you want

The practical upshot

- People almost always WANT the credible reading.
- The Bayesian interval delivers it directly.
- The frequentist interval is routinely misread as it.

When they roughly agree

- With lots of data and a weak prior, they nearly coincide.
- The gap is widest with small samples or strong priors.
- The MEANING differs even when the numbers are close.

The Posterior Predictive

Forecasting the Bayesian way

To forecast, a Bayesian doesn't just plug in a best parameter. The posterior predictive averages the forecast over the ENTIRE posterior, folding parameter uncertainty into the prediction itself.

The idea

- For each possible parameter, form a forecast.
- Weight each forecast by its posterior probability.
- Average them into one predictive distribution.

Why it is better

- It admits we don't know the parameter exactly.
- So the forecast is honestly wider.
- It avoids the false precision of a plug-in.

The Posterior Predictive, Formally

Integrate the forecast over the posterior

$$\text{posterior predictive: } p(y_{\text{new}} \mid \text{data}) = \int p(y_{\text{new}} \mid \theta) p(\theta \mid \text{data}) d\theta$$

Reading it

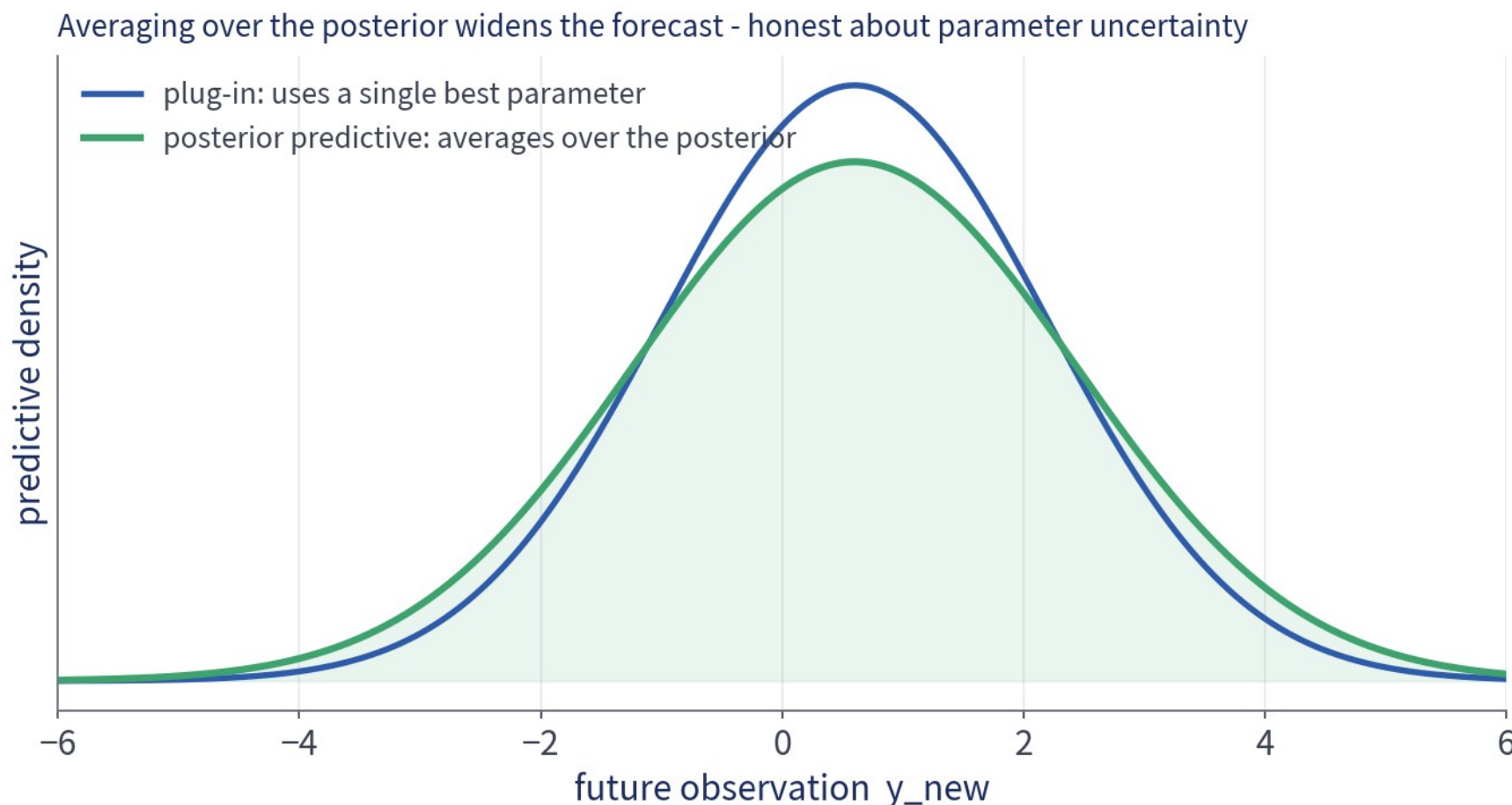
- Inside: the forecast for a fixed parameter.
- Weighted by: the posterior for that parameter.
- Integrated over: all possible parameter values.

The result

- A single predictive distribution for new data.
- Wider than any single-parameter forecast.
- It carries both data noise and parameter uncertainty.

Averaging Widens the Forecast

A plug-in forecast (blue) uses one best parameter and looks confident. The posterior predictive (green) averages over the posterior — wider, with heavier tails, and honest about what we don't know.



Point Forecast vs Predictive Distribution

A number versus a distribution of outcomes

A point forecast

- A single predicted value for the future.
- It hides how uncertain that prediction is.
- It can look far more precise than it is.

A predictive distribution

- A full distribution over future outcomes.
- Its spread reflects true forecast uncertainty.
- You can read off any prediction interval you need.

Why Bayesian Methods Suit Finance

Where the approach earns its keep

Finance is the natural home for Bayes: data is short and noisy, regimes shift, and analysts hold real prior views. Bayesian methods turn all three from problems into features — honest uncertainty, stabilized estimates, and a place for judgment.

The conditions

- Short samples and low signal-to-noise.
- Structural change — the past isn't fully the future.
- Analysts with genuine prior views.

The Bayesian answer

- Honest error bars instead of false precision.
- Priors that stabilize small-sample estimates.
- A principled slot for informed judgment.

Small Samples and Structural Change

Finance rarely gives you clean, abundant data

Why estimates are fragile

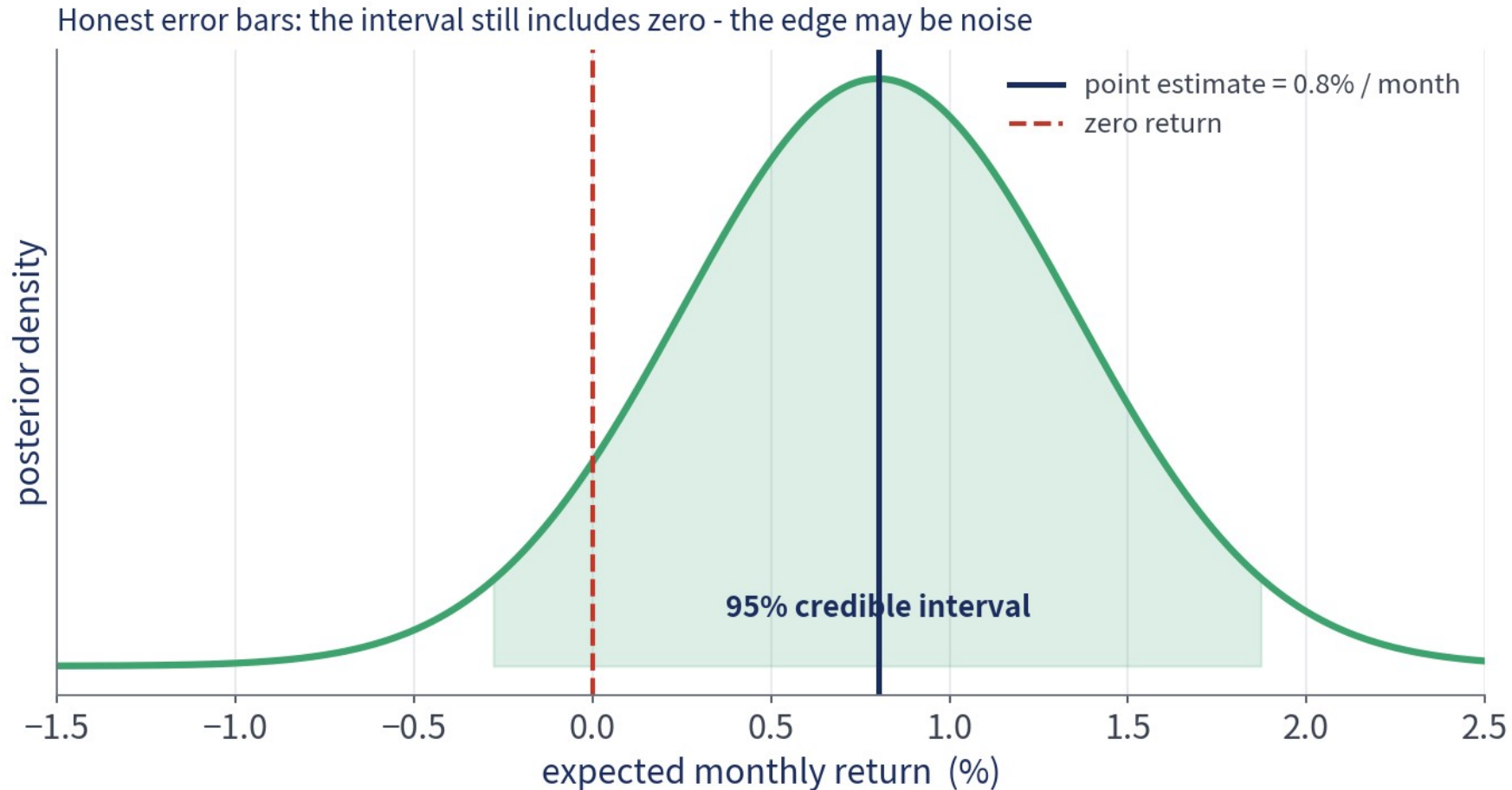
- Return series are short relative to their noise.
- Relationships drift and break over time.
- A point estimate can be badly overconfident.

How Bayes helps

- Priors shrink wild small-sample estimates.
- Posteriors show honestly how little we know.
- You size decisions to real, not assumed, certainty.

Honest Error Bars on a Return Estimate

A posterior for an expected monthly return, from a short sample. The point estimate looks positive, but the 95% credible interval still spans zero — the apparent edge may be noise. That caution is the value.



Frequentist vs Bayesian

Two valid tools, different strengths

Aspect	Frequentist	Bayesian
Parameter	a fixed unknown	a distribution
Main output	point estimate + CI	a full posterior
Interval meaning	about the procedure	about the parameter
Prior	none	explicit
Small samples	can be unstable	stabilized by the prior

Neither is 'right' — they answer different questions. Bayes shines when uncertainty must be explicit.

When to Use Which

Matching method to problem

Lean frequentist when

- Data is abundant and the model is simple.
- You want a quick, standard, defensible estimate.
- No strong prior knowledge is at stake.

Lean Bayesian when

- Data is scarce or the model is complex.
- You have real prior knowledge to include.
- Honest, propagated uncertainty is the point.

The Catch: Computing the Posterior

Why Bayes waited 200 years for computers

Everything rested on one integral — the normalizer. For the Beta-Binomial it was free, thanks to conjugacy. For real models it is intractable, and that single obstacle held Bayesian methods back for two centuries.

The easy cases

- Conjugate models update by arithmetic.
- The normalizer cancels or has a formula.
- No integral to actually compute.

The hard cases

- Realistic models have no conjugate shortcut.
- The normalizer is a high-dimensional integral.
- It cannot be done exactly, by hand or otherwise.

Why Exact Bayes Doesn't Scale

The curse of dimensions

The problem

- The normalizer integrates over every parameter.
- With many parameters, that space is enormous.
- Grid or analytic integration becomes impossible.

The consequence

- Exact posteriors exist only for special models.
- Real financial models are not so lucky.
- We need a way to work with the posterior indirectly.

The Bridge to MCMC

Sample the posterior instead of computing it

The key idea

- Don't compute the posterior — draw SAMPLES from it.
- Samples need only the shape, not the normalizer.
- From samples you get means, intervals, everything.

What makes it possible

- Markov chain Monte Carlo builds such samplers.
- It sidesteps the intractable integral entirely.
- That is exactly next week's subject.

Error Bars on Everything We Built

Every fitted number from this course — AR coefficients, GARCH volatility, a hidden state, a regime probability, rate parameters — can become a posterior. Bayes adds an honest error bar to each; MCMC makes it computable

Error Bars on Everything We Built

Every point estimate from this course can become a posterior distribution.

AR / ARMA

point estimate of the coefficients



GARCH

point estimate of the volatility



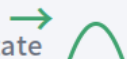
VAR

point estimate of the dynamics



Kalman

point estimate of the hidden state



HMM

point estimate of the regime probs



OU / rates

point estimate of kappa, theta, sigma



The Bayesian upgrade

Wherever we reported a single fitted number, Bayes replaces it with a posterior - a best guess AND an honest error bar. In finance, where data is short and noisy, that uncertainty is not a nicety - it is the point.

The catch: computing these posteriors needs the tools of Week 14 (MCMC).

Credible Intervals and Prediction

Direct Claude Code to read a posterior fully

GOAL From a posterior, extract a credible interval and a posterior predictive, and contrast with a plug-in.

[INPUT] Prompt · Claude Code

Write and run `posterior_use.py` that:

1. takes a posterior (e.g. Beta or Normal) from a Bayesian fit of a probability or a mean return;
2. computes and plots the 95% CREDIBLE interval;
3. builds the POSTERIOR PREDICTIVE for a new observation by averaging over the posterior;
4. overlays a PLUG-IN forecast that uses only the posterior mean, to show it is too narrow.

Report whether the credible interval includes a 'no-edge' value (like zero return).

How to work with it

- The credible interval reads off the posterior.
- The predictive is wider than the plug-in.
- An interval spanning zero is a real warning.
- This is the honest way to forecast.

Deliverable: `posterior_use.py` + a credible interval and a predictive-vs-plug-in comparison.

Three Questions Before You Finish

If you can answer these, Week 13 has landed

Q1	A credible interval is a probability statement about what?	The parameter
Q2	Why is the posterior predictive wider than a plug-in forecast?	It adds parameter uncertainty
Q3	What single obstacle held Bayes back for 200 years?	Computing the normalizer

All three come from the credible interval, the predictive average, and the intractable integral.

Three Pitfalls to Avoid

Where Bayesian practice slips

Common mistakes

- "A confidence interval is a 95% probability for this θ ."
- "A plug-in forecast captures the uncertainty."
- "A positive point estimate means a real edge."

The reality

- That reading is the CREDIBLE interval, not the confidence one.
- Only the predictive folds in parameter uncertainty.
- Check the interval — it may well span zero.

Week 13 in Three Ideas

Pulling the whole week together

1

Bayes updates belief into a posterior

Parts 1-2: posterior proportional to likelihood times prior, made concrete by the Beta-Binomial. A parameter becomes a distribution, updated by data, with conjugacy turning the update into arithmetic.

2

The posterior answers real questions

Part 3: credible intervals give a direct probability about the parameter, and the posterior predictive gives honest forecasts that carry parameter uncertainty — a natural fit for short, noisy financial data.

3

The catch is computation

Real posteriors need an intractable integral. That single obstacle is what Markov chain Monte Carlo overcomes — by sampling the posterior instead of computing it.

Next — Week 14: Markov chain Monte Carlo, the engine that makes Bayesian inference practical.

★ WEEK 13 · PART 3 · PRACTICE

Credible Intervals and Prediction, by Hand

A Direct Probability, and an Honest Forecast

The credible interval and the posterior predictive

Part 3's Bayesian outputs, with the arithmetic shown. We compute a 95% credible interval on an expected return - a direct probability statement - then build a posterior predictive interval that is honestly wider because it carries the estimation uncertainty.

3P

PART 3 · APPLICATIONS

From Posterior to Decision

Case A: a 95% credible interval on an expected return, and how it differs from a confidence interval. Case B: the posterior predictive - a forecast interval that adds parameter uncertainty and is honestly wider than the plug-in. Then a Claude Code lab.

What Part 3 Told Us

Now we compute the outputs - with numbers

1

A credible interval is a direct probability

It says the parameter is in this range with 95% probability, given the data - the statement people actually want.

2

The posterior predictive forecasts honestly

It forecasts the next outcome by integrating over the posterior, so it carries both outcome and estimation uncertainty.

3

Bayes suits finance, at a cost

Direct probabilities and honest forecasts are ideal for decisions - but exact computation doesn't scale, which is why MCMC exists.

Everything here is a calculator exercise: one posterior, one credible interval, one forecast.

How These Cases Work

Data, then arithmetic, then meaning

The shared setup

- Posterior on the mean return: $N(10\%, 8\%)$.
- Year-to-year return volatility: 15%.
- A short history, so real uncertainty.

What you should watch

- How the credible interval reads directly.
- How the predictive adds parameter uncertainty.
- Why the predictive interval is wider.

The Credible Interval

A probability about the parameter

Our posterior on a strategy's expected return is centered at 10% with an 8% spread. From it, a 95% credible interval is a direct statement: there is a 95% probability the true expected return lies in this range, given the data. That's exactly what a decision-maker wants - and it's not what a confidence interval says

The posterior

- Expected return centered at 10%.
- Posterior spread of 8%.
- Reflecting a short track record.

What we'll do by hand

- Take the central 95% of the posterior.
- Read off the interval.
- Contrast it with a confidence interval.

The Credible Interval

Direct probability about the parameter

95% credible interval : $P(a \leq \theta \leq b \mid \text{data}) = 0.95$ (a direct probability about the parameter)

How to read it

- It's a probability about the parameter itself.
- Conditioned on the data you actually saw.
- The natural interval for a decision.

How to build it

- Take the central 95% of the posterior.
- For a Gaussian, mean plus or minus 1.96 sd.
- The endpoints are the interval.

The 95% Credible Interval

Central band of the posterior

$$[a, b] = \mu_{\text{post}} \pm 1.96 \sigma_{\text{post}} = 10\% \pm 1.96(8\%) = [-5.7\%, 25.7\%]$$

The calculation

- 1.96 times the 8% spread is 15.7%.
- Add and subtract from the 10% center.
- The interval is -5.7% to 25.7%.

The reading

- A 95% probability the mean is in this band.
- It's wide - the history is short.
- The band even includes negative returns.

Credible vs Confidence

Two intervals, two meanings

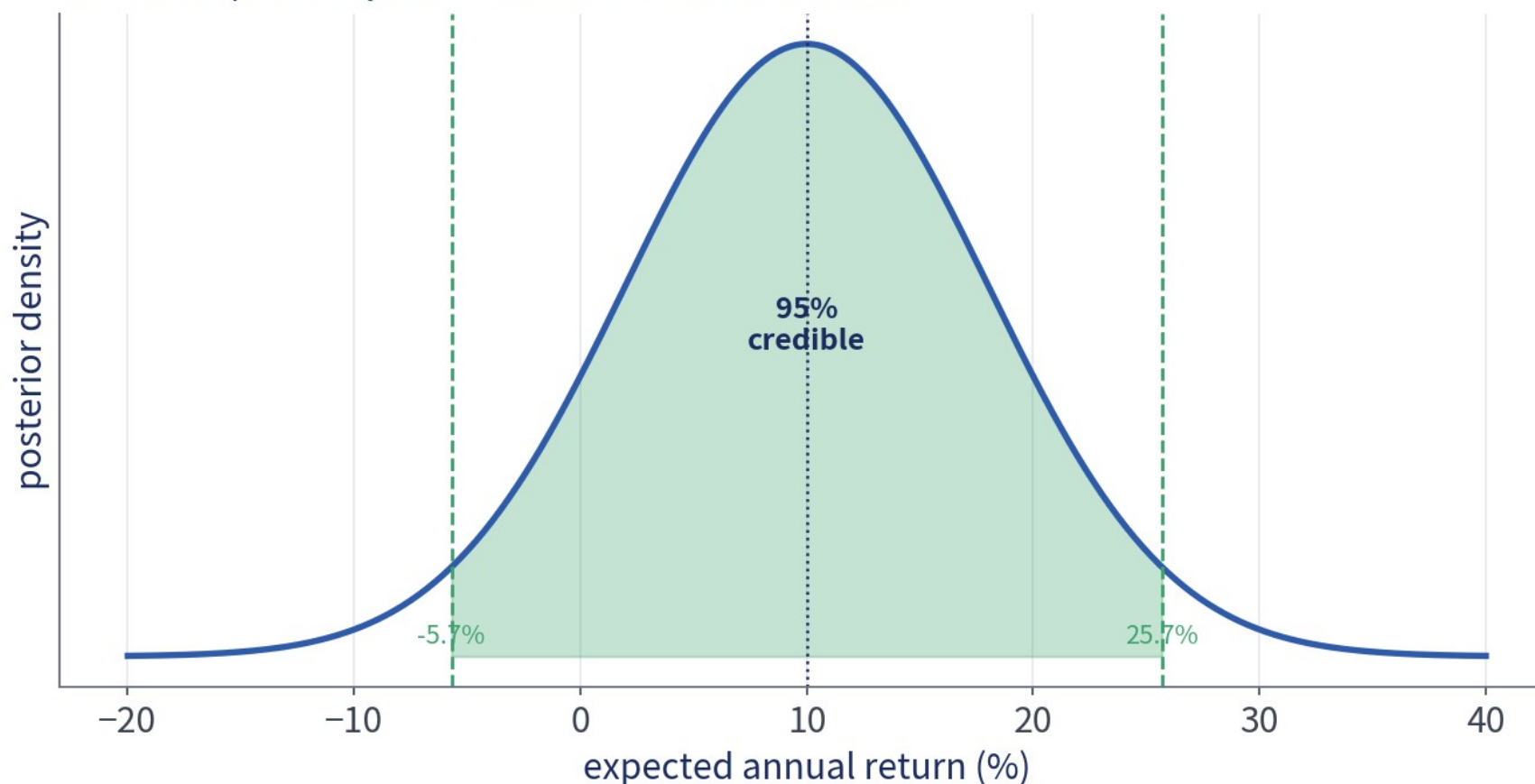
credible: 95% probability $\theta \in [a, b]$ given this data
confidence: 95% of such intervals cover θ (a statement about the procedure)

The credible interval makes a probability statement about the parameter given this data. The confidence interval instead describes the long-run behavior of the procedure - not this specific interval.

The Central 95%

The shaded band holds the central 95% of the posterior. Its interpretation is direct: given the data, there's a 95% probability the true expected return falls between -5.7% and 25.7%. The width reflects genuine uncertainty from the short history

Case A: 95% probability the true mean lies in the shaded band



Why the Distinction Matters

Say what you mean about probability

What people want

- The chance the parameter is in a range.
- A statement about this data, this estimate.
- That's the credible interval.

What confidence gives

- A guarantee about repeated procedures.
- Not a probability about this interval.
- Useful, but a different question.

The Posterior Predictive

Forecasting with honest uncertainty

To forecast next year's return, a naive approach plugs in the 10% estimate and uses the 15% return volatility. But that ignores the uncertainty in the estimate itself. The posterior predictive adds it in: the forecast variance is the outcome variance plus the parameter variance - so the interval is honestly wider. We compute both

The two uncertainties

- Outcome noise: returns vary 15% year to year.
- Parameter uncertainty: the mean's 8% spread.
- Both belong in an honest forecast.

What we'll do by hand

- Add the two variances.
- Take the square root for the predictive sd.
- Compare the interval to the plug-in.

The Posterior Predictive

Outcome noise plus parameter uncertainty

$$\text{Var}(\text{next return}) = \underbrace{\sigma_{\text{data}}^2}_{\text{outcome noise}} + \underbrace{\sigma_{\text{post}}^2}_{\text{parameter uncertainty}}$$

The plug-in error

- Using only the outcome volatility.
- It pretends the mean is known exactly.
- So its interval is too narrow.

The predictive fix

- Integrate the forecast over the posterior.
- That adds the parameter variance.
- The total is the predictive variance.

The Predictive SD

Combine the two spreads

$$\sigma_{\text{pred}} = \sqrt{\sigma_{\text{data}}^2 + \sigma_{\text{post}}^2} = \sqrt{15^2 + 8^2} = \sqrt{289} = \mathbf{17\%}$$

Adding the variances - 15-squared plus 8-squared - gives 289, whose square root is exactly 17%. The predictive spread is bigger than the 15% outcome volatility alone.

Predictive vs Plug-In

How much wider the honest interval is

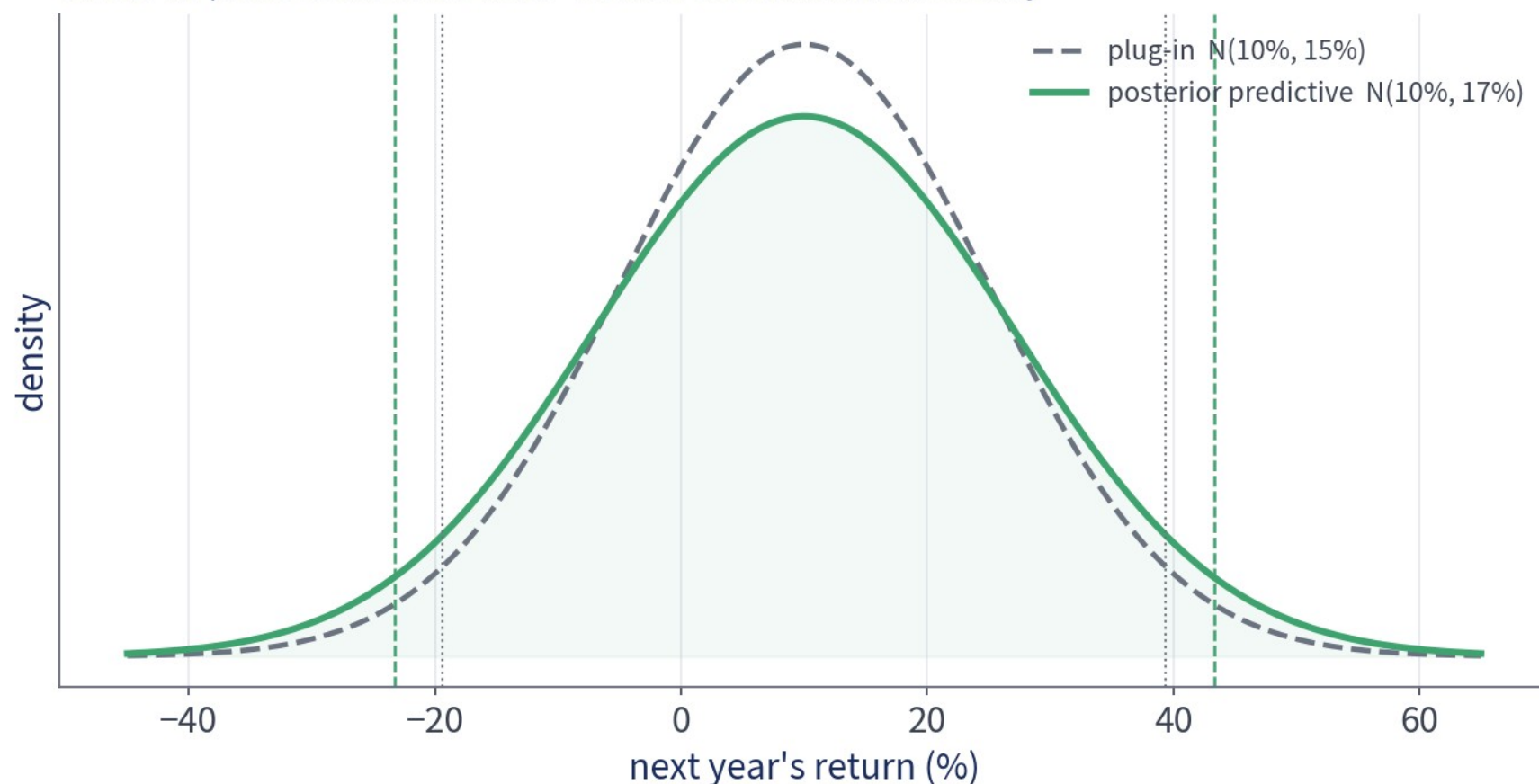
$$\begin{aligned}\text{plug-in: } & 10\% \pm 1.96(15\%) = [-19.4\%, 39.4\%] \\ \text{predictive: } & 10\% \pm 1.96(17\%) = [-\mathbf{23.3\%}, \mathbf{43.3\%}] \text{ (wider)}\end{aligned}$$

The plug-in forecast interval runs -19.4% to 39.4%; the posterior predictive runs -23.3% to 43.3% - wider on both ends, because it admits we don't know the mean exactly.

Honestly Wider

Both forecasts center on 10%, but the posterior predictive (green) has fatter tails than the plug-in (dashed) - carrying the extra uncertainty about the mean. Ignoring that estimation error would leave you overconfident about next year's range of outcomes

Case B: the predictive interval is wider - it carries the estimation uncertainty



Why Honest Intervals Matter

Overconfidence is expensive

The plug-in trap

- Treating the estimated mean as the truth.
- Reporting a forecast interval too narrow.
- Underpricing tail risk.

The predictive fix

- Carry the estimation uncertainty forward.
- Widen the interval accordingly.
- Especially important on short histories.

Why Bayes Suits Finance

And the catch that shapes its history. **Both outputs are what finance actually needs: a direct probability about a parameter, and a forecast that owns its uncertainty. That's why Bayesian methods fit decision-making so well. The catch is computation - exact posteriors don't scale beyond simple models, which is why the modern toolkit leans on MCMC to sample what it**

The strengths

- Credible intervals: direct probability.
- Posterior predictive: honest forecasts.
- Natural for sizing and risk.

in

The catch

- Exact Bayes doesn't scale with dimensions.
- The evidence integral becomes intractable.
- MCMC samples the posterior instead.

Reproduce the Interval and the Forecast

Direct Claude Code to check your arithmetic

GOAL Verify the credible interval and the posterior predictive, then compare to the plug-in.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - credible interval:  
* from posterior  $N(0.10, 0.08)$ , print the 95%  
  credible interval (mean  $\pm 1.96$  sd).  
CASE B - posterior predictive:  
* with outcome vol 0.15 and mean uncertainty 0.08,  
  print the predictive sd =  $\sqrt{0.15^2 + 0.08^2}$ ;  
* print the plug-in and predictive 95% intervals.  
Then plot the posterior with its credible band, and  
the plug-in vs predictive densities.  
Match every printed number to the slides.
```

Check against the slides

- credible: [-5.7%, 25.7%].
- predictive sd = 17%.
- plug-in: [-19.4%, 39.4%].
- predictive: [-23.3%, 43.3%].

Deliverable: part3_practice.py whose printed numbers match the credible interval and the forecasts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does a credible interval say?

95% probability the parameter is in the range

Q2 Why is the predictive wider than the plug-in?

It adds the parameter uncertainty

Q3 Why does exact Bayes need MCMC?

The posterior doesn't scale with dimensions

All three come from the credible interval and the posterior predictive.

Three Pitfalls to Avoid

Where the outputs trip people

Common mistakes

- "A confidence interval gives the probability here."
- "Plug in the estimate and use outcome vol."
- "Exact Bayes works for any model."

The reality

- Only the credible interval is a direct probability.
- The predictive adds parameter uncertainty - it's wider.
- Beyond simple models, you need MCMC to sample it.

Two Outputs to Remember

Part 3 applications in one screen

1

The credible interval is a direct probability

From a posterior of $N(10\%, 8\%)$, the 95% credible interval is -5.7% to 25.7% - a direct statement that the true mean is in that band with 95% probability, given the data. A confidence interval, by contrast, describes only the procedure.

2

The posterior predictive is honestly wider

Forecasting next year's return, the predictive sd combines the 15% outcome volatility with the 8% parameter uncertainty into 17%, widening the interval from -19.4%-39.4% (plug-in) to -23.3%-43.3%. Ignoring estimation error breeds overconfidence.

3

Bayes fits finance, MCMC makes it scale

Direct probabilities and honest forecasts are exactly what decisions need. But exact posteriors don't scale, so the modern toolkit samples them with MCMC - the natural next step.

Next — Week 14: sampling the posterior when it has no closed form, with Markov chain Monte Carlo.